

Review

Recent advances in flexible bioelectronics and intelligent wearable devices for wound management

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SUMMARY

Large, non-healing, or chronic wounds are a global medical challenge that affects millions of patients. Currently, the effectiveness of clinical wound management is still limited by passive treatment strategies, which typically rely on physicians' experience and laboratory tests for qualitative assessments, resulting in prolonged hospitalization, increased patient suffering, and even more severe complications. In recent years, with the rapid development of flexible electronics and wearable devices, various types of bioelectronic dressings/bandages have been extensively explored and applied in wound management. This review aims to systematically review the latest advances in flexible electronics and intelligent wearable devices for wound monitoring and treatment. First, the progress of wound exudate management technology is introduced. Second, sensing of wound biomarkers, including material innovation, sensing mechanisms, and performance advantages, is summarized. Third, diverse therapeutic strategies integrated into wearable devices are reviewed, emphasizing the targeted and on-demand intervention capabilities. Fourth, integrated closed-loop systems that combine power supply, artificial intelligence assistance, and communication are discussed. Finally, the current challenges and future prospects are discussed. This review aims to provide comprehensive insights for researchers and clinicians engaged in the development and application of flexible electronics and intelligent wearable devices for wound care.

INTRODUCTION

Normal wound healing goes through four stages: hemostasis, inflammation, proliferation, and remodeling.¹ During each stage, different cells are recruited and activated and migrate and proliferate to repair damaged tissue. But this carefully orchestrated process is prone to being impaired by various comorbidities, such as diabetes, infections, ischemia, metabolic disorders, immunosuppression, and radiation. In such cases, the healing wound microenvironment will be affected or even destroyed, leading to a poor prognosis.² In the US alone, there are over 8 million patients suffering from chronic wounds, with an annual economic impact exceeding \$30 billion.³ Most commercially available dressings are passive and lack objective quantitative indicators to offer insights into the wound status, which hinders healthcare providers from making accurate diagnoses and

timely interventions, potentially leading to the occurrence of chronic wounds and infections.

Real-time acquisition of key indicators of the wound microenvironment is of great significance for accurately assessing the healing stage and providing timely treatment. The significant progress made in bioelectronics, involving materials science, biomedicine, and electronic engineering, has brought revolutionary breakthroughs to the field of wound management. Flexible and wearable electronics are some of the main focuses of bioelectronics. By converting biological signals into electrical signals and transmitting them to external devices, objective quantitative information can be obtained, and interventions may be administered when necessary through flexible and wearable electronics, which not only form a closed-loop mode of sensing, transmission, and treatment but also promote the transformation of wound management from a static and passive mode to a dynamic and active mode.^{4–9} For



improved wound healing outcomes, an ideal flexible and wearable electronic dressing platform needs to meet the following requirements: (1) flexibility, portability, and wireless operation to avoid any undesired tethering and discomfort caused by conventional rigid devices; (2) integration with sensing or treatment modalities (autonomous and closed-loop management would be desirable); (3) a well-designed bioelectronic interface for conformal contact with skin to improve patient comfort and compliance,^{10–13} stable signal transduction and intervention, and on-demand adhesion to avoid possible secondary damage when removal; (4) gas permeability and the ability to manage wound exudate efficiently to maintain optimal moisture; and (5) good biocompatibility and antibacterial properties and the ability to promote wound healing, such as promoting angiogenesis and re-epithelialization, reducing scar formation.

This review provides an overview of recent progress in wearable bioelectronic devices for wound management. We summarize efforts to construct wearable platforms to improve personalized wound assessment and treatment, ranging from wound exudate management to wound status sensing as well as treatment. We also present integrated closed-loop systems combining a power supply mode, AI algorithm assistance, and communication modules. Finally, challenges and future prospects for applying wearable bioelectronic devices for wound care are discussed.

SENSING AND MONITORING OF WOUND STATUS

The development of sensing systems is driving the shift of signal monitoring from offline, isolated, and *ex vivo* to *in situ*, continuous, and real time. Sensing systems are typically constructed by building functional electrodes on flexible substrates, mainly involving the use of materials or biological feature elements that can respond to biological signals, selectively capturing target analytes, and converting biological signals into readable electrical or optical signals through an analog-to-digital converter. A variety of flexible bioelectronic sensors have been developed for diverse monitoring targets, with representative studies and their key features summarized in Table 1.

Wound exudate management

Wound exudate is produced during the normal process of wound healing to maintain moisture balance and facilitate nutrient delivery and immunomodulation. In cases of wound infection, tissue necrosis, inadequate perfusion, and other adverse events, the components of wound exudate can become abnormal. The composition of exudate varies depending on the wound condition. Therefore, effective collection and analysis of wound exudate is of great significance to achieve early warning and timely treatment. However, the exudate rate of wound fluid is very slow, only $1\text{--}10\ \mu\text{L cm}^{-2}\ \text{h}^{-1}$, which makes its collection and analysis challenging. Moreover, due to the complex matrix of wound exudate, it is difficult to develop sensors that can accurately perform *in situ* analysis.

A microfluidic module is commonly used to collect wound exudate.^{14,20,26} For example, Ge et al. developed a smart dressing with Janus-modified polydimethylsiloxane (PDMS) microfluidic channels, enabling unidirectional self-pumping of wound exudate to a spiral storage channel. The asymmetric wettability

design prevents exudate backflow, isolating the wound from excess fluid, creating a favorable environment, and reducing the infection risk (Figure 1A).²⁶ In addition, in order to distinguish between fresh and old exudate, Wang et al. developed a wearable microfluidic device named iCares that incorporates a pump-free triad of microfluidic modules, including a Janus membrane, wedge channels, and graded micropillars, to enable unidirectional collection and refreshing of low-volume wound exudate (Figure 1B). This design ensures efficient management of inherently limited wound exudate and continuous *in situ* of biomarkers.²⁰ Deng et al. developed a biomimetic microfluidic module inspired by tomato glandular trichomes (Figures 1C and 1D), incorporating an asymmetric wedge-shaped pattern and super-contrast wettability surface engineering to achieve directional, efficient, and skin-conformal collection and transport of wound exudate within 0.55 s, enabling real-time analysis without backflow.¹⁴

Traditional analytical methods rely on sampling inherently limited wound exudate and then conducting laboratory analysis, which is a labor-intensive, time-consuming process, and it is difficult to capture these transient reactive species. Recent advances in wearable electronics-based wound exudate management platforms have shown promise because of their ability to collect, sample, and analyze *in situ*. It still remains a challenge to effectively collect wound exudate *in situ* and control its collection rate, which may otherwise lead to problems such as inconsistent sample concentrations and unreliable signals. Furthermore, the wound exudate contains a variety of complex substances, which may affect the detection accuracy of certain target substances. Therefore, using simulated wound fluid alone to detect target substances may result in deviations from the actual results, making it necessary to use real wound exudate and perform *in vivo* experiments for validation in the future.

Physical signal sensors

Temperature sensor

Temperature is considered one of the classic signs during wound healing. Generally, increased temperature is associated with inflammation and infection, while decreased temperature indicates a delayed healing process due to inhibited metabolic rate.

Electrical temperature sensors are mainstream-type temperature sensors, such as resistance type and capacitance type, among which the resistance type is the most commonly used. Its principle is based on the resistance change of materials, such as metals (e.g., gold, nickel, copper, and platinum),^{30–34} nanomesh/nanowires,³⁵ semiconductors (e.g., graphene and silicon),^{27,31,36,37} and carbon²⁰ or a thermistor. For example, Deng et al. fabricated a compact, serpentine-line structural, laser-engraved graphene (LEG) temperature sensor by engraving the flexible polyimide film using a CO₂ laser. As temperature rises, the sensor's conductivity increases owing to increased electron-phonon scattering and the thermal velocity of electrons in the sandwiched layers (Figure 1E), which results in a linear correlation between resistance variation and temperature variation over the range of 30°C–50°C with a sensitivity of approximately $-0.056\% \text{ } ^\circ\text{C}^{-1}$ (Figure 1F).¹⁴ Jiang et al. integrated a thermistor for temperature monitoring to warn of infection-related

Table 1. Representative studies on flexible bioelectronic sensors for sensing

Representative study	Sensing modality	Analytes	Validation level	Treatment	Key advantages	Key limitations	Translational potential
Deng et al. ¹⁴	physical and biochemical signals	O ₂ temperature UA lactate pH	<i>in vitro</i> , animal model	no treatment	high sensitivity (–0.056% °C ^{–1}), linear range 30°C–50°C for temperature	difficulty in distinguishing wound from surrounding skin	high; it can be integrated with microfluidic platforms
Kang et al. ¹⁵	physical signal	temperature	<i>in vitro</i> , animal model	thermotherapy	large-area thermal mapping, resolution of 0.3°C	complex fabrication, requiring shielding layer	medium; it can be suitable for spatial temperature distribution
Jiang et al. ¹⁶	physical signals	impedance temperature	<i>in vitro</i> , animal model	electrical stimulation	detecting healing progression and early infection	relationship between impedance and wound state needs further elucidation.	high; it can be integrated into wireless closed-loop system
Ge et al. ¹⁷	physical signal	pressure	<i>in vitro</i> , animal model	no treatment	real-time plantar pressure mapping, threshold alert (high sensitivities of 3.356 V N ^{–1} under 0.01–0.5 N pressures)	limited linear range, reduced sensitivity under high pressure (sensitivities of 2.524 V N ^{–1} under high (0.5–10 N) pressures)	high; it can be suitable for diabetic foot ulcer prevention
Montazerian et al. ¹⁸	biochemical signal	pH	<i>in vitro</i> , animal model	no treatment	~250% improved pH sensitivity, infection detection	complex fabrication, limited long-term stability	medium; it can be suitable for injectable conductive hydrogel platforms
Liu et al. ¹⁹	physical and biochemical signals	sodium potassium calcium pH UA temperature	<i>in vitro</i> , animal model	no treatment	an integrated multiplex sensing bandage system capable of assessing early wound infection and inflammation, linear range of 0–1000 μM, high selectivity for UA	enzyme activity susceptible to environmental conditions	high; it is suitable for early infection monitoring
Wang et al. ²⁰	physical and biochemical signals	NO H ₂ O ₂ pH temperature	<i>in vitro</i> , animal model, clinical trial	no treatment	linear range of 0–45 μM, LOD of 0.5 μM for NO sensitivity of 5.91 nA/μM, LOD of 0.25 mM for H ₂ O ₂	requiring cold storage of Hb, limited long-term stability	high; it is integrated into the iCares microfluidic system

(Continued on next page)

Table 1. Continued

Representative study	Sensing modality	Analytes	Validation level	Treatment	Key advantages	Key limitations	Translational potential
Garland et al. ²¹	biochemical signal	lactate	<i>in vitro</i> , animal model	no treatment	wireless, battery-free, predicts wound closure rate	oxygen deficiency may affect detection accuracy	medium; it requires pH calibration
Zheng et al. ²²	biochemical signal	glucose pH Urea UA total protein	<i>in vitro</i>	no treatment	linear range of 0–5 mM, LOD 0.27 mM for glucose	semi-quantitative, relies on image analysis	medium; it can be suitable for visually assisted monitoring
Chen et al. ²³	macromolecular signal	IL-6	<i>in vitro</i> , animal model, clinical trial	no treatment	LOD as low as 0.047 pg/mL, accuracy 99.8%	complex system, requires neural network assistance	high; it is suitable for diabetic wound exudate
Deng et al. ²⁴	macromolecular signal	MMP-9	<i>in vitro</i> , animal model, clinical trial	stimulus-responsive drug delivery	linear range of 0.01–25 µg/mL, LOD 8.9 ng/mL	not yet validated in animal	medium; it can be integrated with antibacterial drug release
Simoska et al. ²⁵	macromolecular signal	PYO	<i>in vitro</i> , cell culture	no treatment	linear range of 1–250 µM, LOD 1.0 µM	not validated in animal and clinical settings	medium; it can be suitable for rapid detection of <i>P. aeruginosa</i>

temperature abnormalities.¹⁶ The iCares integrated a resistive ink-jet-printed carbon temperature sensor, which detected temperature elevation within 24 h after bacterial inoculation and trended back toward normal after treatment.²⁰ However, it is difficult to distinguish temperature values at the wound site from adjacent skin. Moreover, the temperature values may also vary in different areas. Temperature mapping may help to advance the applicability of temperature sensors to provide more sensing sites and higher resolution across wound areas and to perform precise spatiotemporal mapping over large scales. Liu et al. developed an ultrathin amorphous silicon-based temperature sensor array consisting of 64 resistors for monitoring localized wound inflammation (Figures 1G and 1H), enabling spatial temperature mapping at submillimeter resolution and high sensitivity of 0.1°C. Such an array matrix offers a yield of 100% and a mean per-channel temperature coefficient of resistance of −2.25%, which is an order of magnitude larger than those of noble metal materials.²⁷

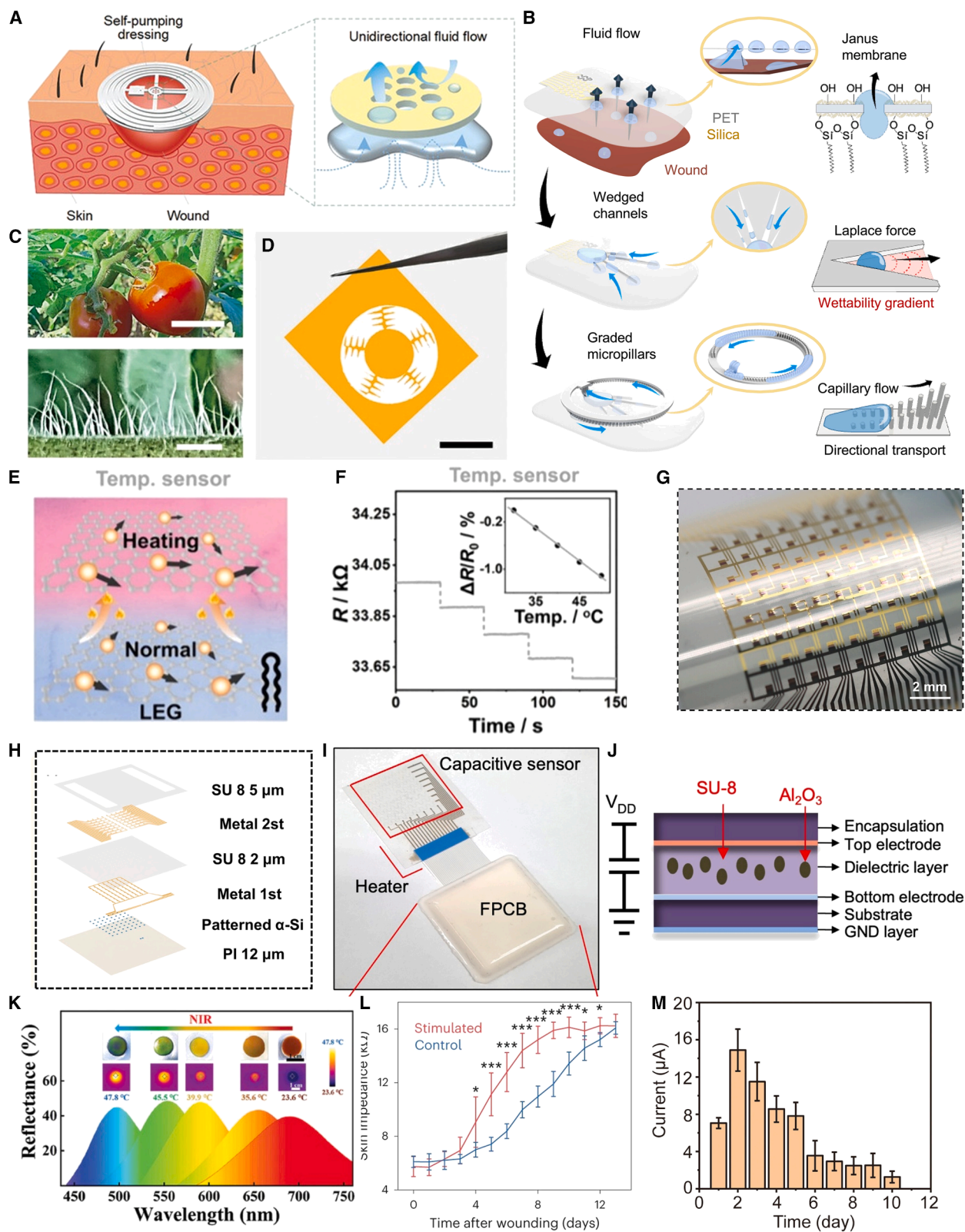
In addition to resistance change, temperature changes can also cause alterations in the dielectric constant of the capacitive medium or the spacing between electrodes in capacitive sensors. For example, Kang et al. developed a wearable graphene-based thermal patch integrated with an 8 × 8 capacitive temperature sensor array (Figure 1I). Fabricated using chemical vapor deposition-grown graphene as electrodes and an SU-8/Al₂O₃ composite as the dielectric layer (Figure 1J), the sensor enables a high resolution of 0.3°C, high sensitivity of 40.4 fF/°C in the range of human body temperature, and the capability of large-area monitoring (3 × 3 cm²) for capturing spatial thermal gradients in wounds. A graphene ground shielding layer further

minimizes noise interference from skin conductivity and sweat, ensuring stable measurements during prolonged wear. The sensor successfully detected a 0.7°C temperature drop and tracked dynamic temperature changes *in vivo*.¹⁵

Additionally, an optical temperature response mechanism through intuitive visual observation or spectral changes can be used. Ji et al. developed a multifunctional hydrogel electronic (MHE) platform based on MXene@metal-organic framework (MOF)/Fe₃O₄@C photonic crystal hydrogel to enable visual temperature monitoring. Under 808 nm near-infrared (NIR) irradiation for photothermally enhanced chemodynamic therapy (prothrombin time/CDT), MHEs exhibit distinct structural color changes (reflection peak shifting from 689 to 497 nm) corresponding to temperature elevation (23.6°C–47.8°C) (Figure 1K). This color-responsive sensing allows real-time visual tracking of therapeutic temperature without additional equipment.²⁸ However, it has an overly broad temperature range, which makes it unable to accurately detect the subtle temperature changes in wound infections. To overcome this limitation, Li et al. reported a flexible colorimetric temperature sensor based on hyperbranched thermosensitive polymer-gold nanoparticles (NPs) for sensing subtle temperature changes. When sensing a temperature rise to the 32°C–34°C range, the polymer contraction alters the aggregation state of AuNPs, causing their color to shift from transparent pink to opaque purple.³⁸

Impedance sensor

During the inflammation phase, wound exudate forms and creates a moist and electrically conductive environment, and the local skin impedance decreases.³⁹ As the wound gradually heals, the wound gradually dries, resulting in increased



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impedance.³ Kekonen et al. developed a screen-printed multi-electrode sensor array for quasi-monopolar bioimpedance measurement. The system tracked impedance changes at multiple frequencies (150 Hz to 5 kHz) for 142 h beneath dressings, finding that the wound impedance gradually recovers as it is healing, reaching the impedance of intact skin at higher frequencies and increasing significantly at the lowest frequency.⁴⁰ When a wound becomes infected, the microbial biofilm produced by bacteria will lead to decreased impedance.⁴¹ Jiang et al. developed a smart bandage to monitor wound impedance change. In the treatment group, a significant increase of wound impedance was observed to reach the plateau faster (Figure 1L). The platform can monitor decreased impedance with infection-induced biofilm formation. It can also detect wound healing progression and early onset of infection, combined with temperature monitoring in animal studies.¹⁶

Song et al. developed a bioresorbable system that integrates impedance sensing with monitoring. The system employs serpentine-structured molybdenum (Mo) electrodes (inner on wound center, outer on perimeter) to reflect impedance changes by measuring current signals that correlate with wound exudate and drying status; current peaks ($\sim 20 \mu\text{A}$) in early inflammation and declines to 0 as the wound heals (Figure 1M), providing a simple yet effective proxy for tracking the healing status.²⁹ However, impedance measurement may vary by device, so methods need to be further standardized.

Pressure sensors

Pressure sensors can provide timely alerts when excessive mechanical stress is applied, reminding the user to relieve local high pressure, which is crucial for pressure ulcer prevention and treatment. The sacrum and sole are prone to pressure ulcers, and skin-prosthetic interfaces may also develop ulcers due to excessive local pressure. Conservatively, the lifetime incidence of foot ulcers in diabetic patients is estimated at 15%–25% and may be even higher, reaching 19%–34%.^{42,43} Since sustained static high pressure is one of the main risk factors for pressure ulcers, it is possible to prevent and reduce the recurrence of ulcers by monitoring local pressure and preventing sustained static high pressure.^{44,45} Generally, the principle of pressure measurement

is typically to convert pressure into electrical signals, such as capacitive,⁴⁶ resistive,⁴⁷ piezoelectric,¹⁷ and triboelectric pressure sensors. Both piezoresistive and capacitive sensors achieve signal conversion by causing the sensitive element to deform under external force, changing its resistance or capacitance, while piezoelectric sensors achieve pressure sensing by detecting the charges.

Capacitive pressure sensors possess the advantages of a simple structure, low power consumption, and easy large-area fabrication.⁴⁸ For instance, Tao et al. fabricated a smart insole system composed of an array of 24 capacitive pressure sensors fabricated by vertically introducing a porous structure into an elastic dielectric layer, capable of mapping both static and dynamic plantar pressure distributions (Figure 2A). Each sensor features a hexagonal configuration with a pore diameter of 600 μm and exhibits a good linear response over a wide detection range of 0–200 kPa with a sensitivity of $12 \times 10^{-3} \text{ kPa}^{-1}$. Figure 2B displays plantar pressure heatmaps for three standing postures.⁴⁶

Piezoresistive pressure sensors are also widely developed due to the ease of fabrication and simple operation processes. Ge et al. developed a dual-layer liquid metal fiber for real-time plantar pressure visualization, threshold-based alerts, and on-demand electrical stimulation (ES) treatment (Figure 2C). The hollow PDMS microtube filled with liquid metal enables pressure-responsive resistance changes (Figure 2D). Figures 2E and 2F presents the right-foot plantar pressure distribution maps and corresponding photographs for various static postures.⁴⁷ However, since some piezoresistive sensors demonstrate a negative correlation with external pressure,⁵² the range of resistance change would be limited to less than 100%,⁵³ which results in the sensor's linearity and sensitivity performing well only under low pressure,^{54,55} restricting the operating range.

In addition to preventing ulcers, pressure sensors can also serve as evaluation and monitoring tools for compliance and adequacy of compression therapy.⁵⁶ Compression therapy is the most commonly used clinical treatment method for venous ulcers. Its therapeutic efficacy relies on precise pressure control.

Figure 1. Wound exudate management

- (A) Schematic of the Janus dressing removing excess wound exudate. Reproduced, with permission, from Ge et al.²⁶ Copyright 2023, Wiley.
- (B) Design of the triad microfluidic modules incorporating the Janus membrane, wedged channels, and graded micropillars for efficient collection, transfer, and refreshing of wound exudate. From Wang et al.²⁰ Reprinted with permission from AAAS.
- (C) Photographs of the tomato (*Solanum lycopersicum*) plant (top; scale bar, 5 cm) and glandular trichomes on its stem surface (bottom; scale bar, 2 mm).
- (D) Photograph of the microfluidics module on the polyimide (PI) layer. Scale bar, 8 mm.
- (E) Schematic of temperature sensing based on LEG.
- (F) Resistance responses of the temperature sensor at different temperatures. Inset: the relevant calibration chart. Reproduced, with permission, from Deng et al.¹⁴ Copyright 2025, American Chemical Society.
- (G) Photograph of the temperature sensor array attached to a curving surface.
- (H) Exploded-view illustration of the temperature sensor layer. Reproduced, with permission, from Liu et al.²⁷ Copyright 2024, National Academy of Sciences.
- (I) Photograph of the 8×8 array capacitive sensor system. Scale bar, 1 cm.
- (J) Cross-sectional view of the capacitive sensor structure. Encapsulation: SU-8, 2 μm ; top and bottom electrodes: two-layer graphene; dielectric layer: SU-8/Al₂O₃ composite 1.5 μm ; substrate: SU-8 3 μm ; and ground (GND): three-layer graphene. Reproduced from Kang et al.¹⁵ under a CC BY-NC 4.0 license.
- (K) Thermal images, structural color transformation images, and corresponding reflection spectra of MHE during NIR irradiation. Reproduced, with permission, from Ji et al.²⁸ Copyright 2025, Wiley.
- (L) Impedance of excisional wounds over time, indicating accelerated tissue regeneration with stimulation. $n = 5$ for each group. All data are presented as mean $\pm$ SD. Two-tailed t test assuming equal variances was performed. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Reproduced, with permission, from Jiang et al.¹⁶ Copyright 2023, Springer Nature.
- (M) Average detected current measurements correlate with wound closure in diabetic animals. Reproduced from Song et al.²⁹ under a CC BY 4.0 license.

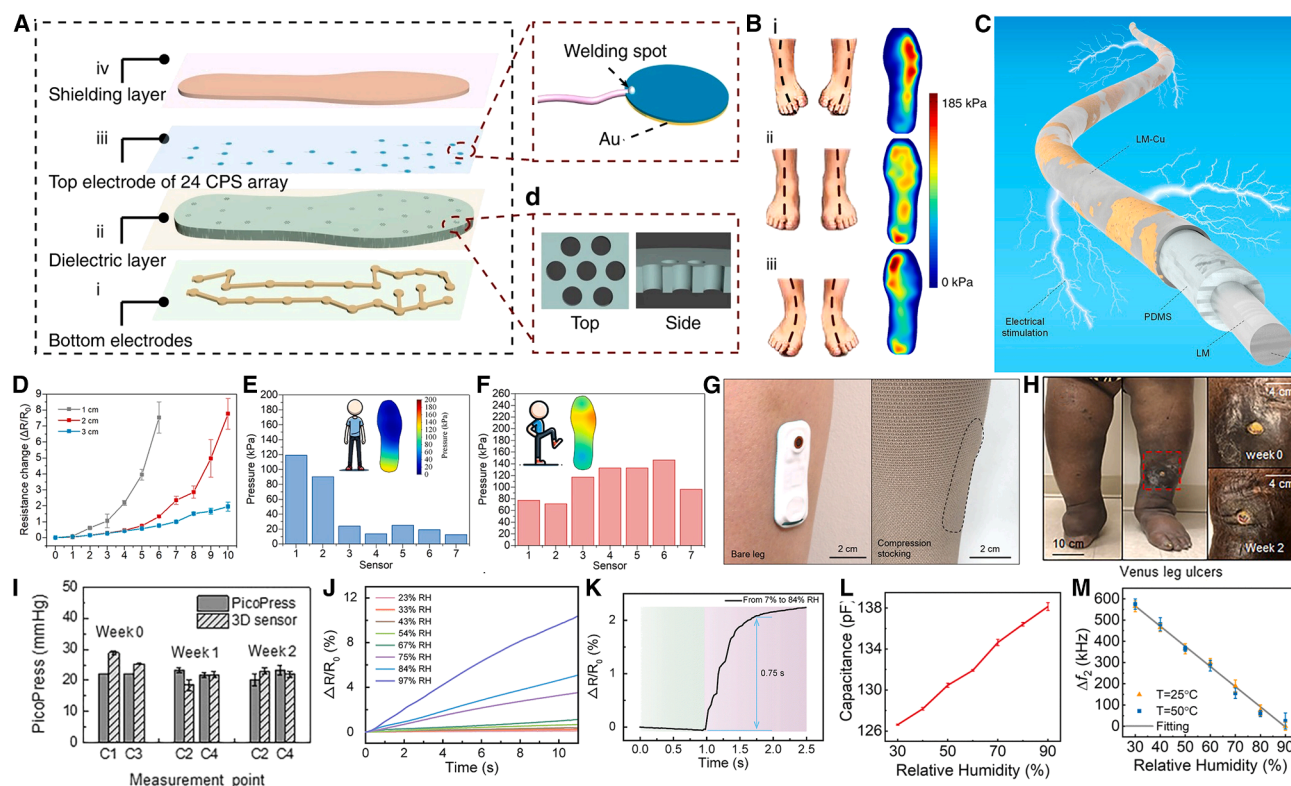


Figure 2. Physical signal sensors

(A) Basic structure of the insole consisting of four layers: i, bottom electrodes connected to the ground; ii, elastic dielectric layer with a 24-channel structured capacitive dielectric layer; iii, top electrodes of the 24 capacitive sensors; and iv, shielding layer connected to the ACS port. Top right inset: illustration of the flexible screen-printed top electrodes with a welding hole. Bottom right inset: top and side view diagrams of the vertical pores.

(B) Three standing postures corresponding to underpronation (i), normal posture (ii), and overpronation (iii) and thermodynamic diagram of the plantar pressure for the three standing postures. Reproduced, with permission, from Tao et al.⁴⁶ Copyright 2020, Springer Nature.

(C) Structural illustration of the flexible LM fiber comprising an LM core, hollow PDMS microtube, and LM-copper conductive coating.

(D) The relationship between the normalized resistance change and applied pressure for fibers of varying lengths.

(E and F) Measured pressures from seven channels on the right plantar surface for four standing postures: double-leg stance (E) and single-leg stance (F). The insets display schematics of the postures (left) and corresponding plantar pressure heatmaps (right). Reproduced, with permission, from Ge et al.⁴⁷ Copyright 2025, Wiley.

(G) Photographs of SCV on the leg of a healthy patient (left) and with a compression stocking (right).

(H and I) Photographs of the left leg and the VLU of a patient (H) and measured pressures for 2 weeks (I). Reproduced from Park et al.⁴⁹ under a CC BY-NC 4.0 license.

(J) The slope of the change in the relative resistance of the coaxial fiber in different RH environments.

(K) The response time of the coaxial fiber from the 7% RH to 84% RH environment. Reproduced, with permission, from Dong et al.⁵⁰ Copyright 2024, Royal Society of Chemistry.

(L) Measured capacitance of the capacitive humidity sensor versus RH.

(M) Frequency shift Δf_2 as a function of RH. Reproduced, with permission, from Ye et al.⁵¹ Copyright 2024, IEEE.

Applying a constant, consistent pressure of ~ 40 mmHg across the skin surface can improve healing rates.^{57,58} However, pressure fluctuates with body movement and bandage loosening. Mehmood et al. developed a sensing system with a piezo-resistive pressure sensor, realizing real-time monitoring of sub-bandage pressure. This system can capture pressure fluctuations; a persistent pressure rise may signal wound infection/limb swelling, while a gradual drop means lost therapeutic pressure.⁵⁹ Similarly, Park et al. developed a skin-interfaced flexible pressure-temperature sensor for compression therapy of venous leg ulcers. It enables real-time, continuous measurement of skin-garment interface pressure to maintain effective thera-

peutic pressure levels (Figure 2G). During a month of clinical study, the venous leg ulcers on the left leg of the patient recovered within 3 weeks, and the pressures measured every week, using a reference and a biomedical sensor for monitoring compression therapy of venous leg ulcers (SCV), exhibited good agreement (Figures 2H and 2I). This result provides evidence of the accurate measurement of interface pressure by SCV in the process of recovery from ulcers. Temperature sensing can be combined with pressure to reflect local inflammation and repair status.⁴⁹ In addition, negative-pressure wound therapy is an effective strategy for the management of contaminated wounds. Monitoring the pressure of wounds under

negative pressure conditions can help to achieve parameter visualization, assisting in the treatment of wound healing.⁶⁰

It is noted that pressure cannot directly characterize the healing speed, nor can it directly indicate the healing stage and reflect tissue repair, but it is a quantitative indicator of the effectiveness of compression therapy and a monitoring parameter for wound healing. Despite this, the integration of pressure sensors into wound monitoring systems provides a way to monitor therapeutic pressure, enriching physical markers for wound management. Future research needs to optimize sensor accuracy, conduct clinical trials on patients, and expand the adaptive scope to promote clinical translation.

Humidity/moisture sensors

Humidity is a key microenvironment parameter for wound healing. Excessive accumulation of exudates and an overly dry environment are both detrimental to tissue repair. Therefore, maintaining appropriate humidity is critical for wound care. In the field of wound monitoring, the current main types of humidity/moisture sensors are electrically based and utilize changes of humidity to induce alterations in the electrical properties of materials.

Carbon nanotubes (CNTs) and graphene are carbon nanomaterials that are often used based on superior electrical conductivity, flexibility, and moisture sensitivity.^{61,62} For example, Li et al. developed a moisture-sensitive e-skin based on a CNT/graphene/gelatin methacryloyl (GelMA) mat. As a moisture sensor, it possesses a high sensitivity coefficient of 12.05, showing a linear relationship between resistance change rate and wetness in the range of 0–14 $\mu\text{L cm}^{-2}$ with a sensitivity coefficient of 12.05 ($R^2 = 0.988$). Notably, it demonstrates strong anti-interference ability against ambient relative humidity (RH) (11.31%–94.62%) and body temperature (35°C–42°C). Integrated with a portable device, this mat successfully realized real-time monitoring of interstitial fluid outflow in rat abdominal wound models.⁶³ MXene also displays high moisture-sensitivity. Dong et al. developed a multifunctional coaxial fiber with a sodium alginate/gelatin/silica NP core and an MXene/silver nanowire (AgNW) sheath, integrating humidity sensing and moisture desorption. This fiber exhibits excellent humidity sensitivity, with relative resistance change ranging from 0.1% to 12% as RH increases from 23% to 97% (Figure 2J) and a rapid response time of only 0.75 s when switching from 7% to 84% RH (Figure 2K). It maintains stable humidity sensing performance over 100 RH cycle tests and 30 days of air exposure.⁵⁰

Capacitance change can also be used. Ye et al. proposed a multimodal wound monitoring system based on higher-order parity time (PT)/PT-reciprocal scaling symmetry, integrating a capacitive humidity sensor into the PT-symmetric resistor-inductor-capacitor (RLC) tank. The increase in capacitance results in a downshifting of f_2 . Operating in the RH range of 30%–90%, this sensor exhibits a high sensitivity of 9.3 kHz/%RH (Figures 2L and 2M). Notably, its humidity-sensing performance is barely affected by temperature variations or the presence of biological tissues, and it can be synchronously detected with temperature, pressure, and pH without crosstalk.⁵¹

Bottlenecks

Despite advances in the flexibility of physical signal sensors and array integration, there are some challenges regarding ap-

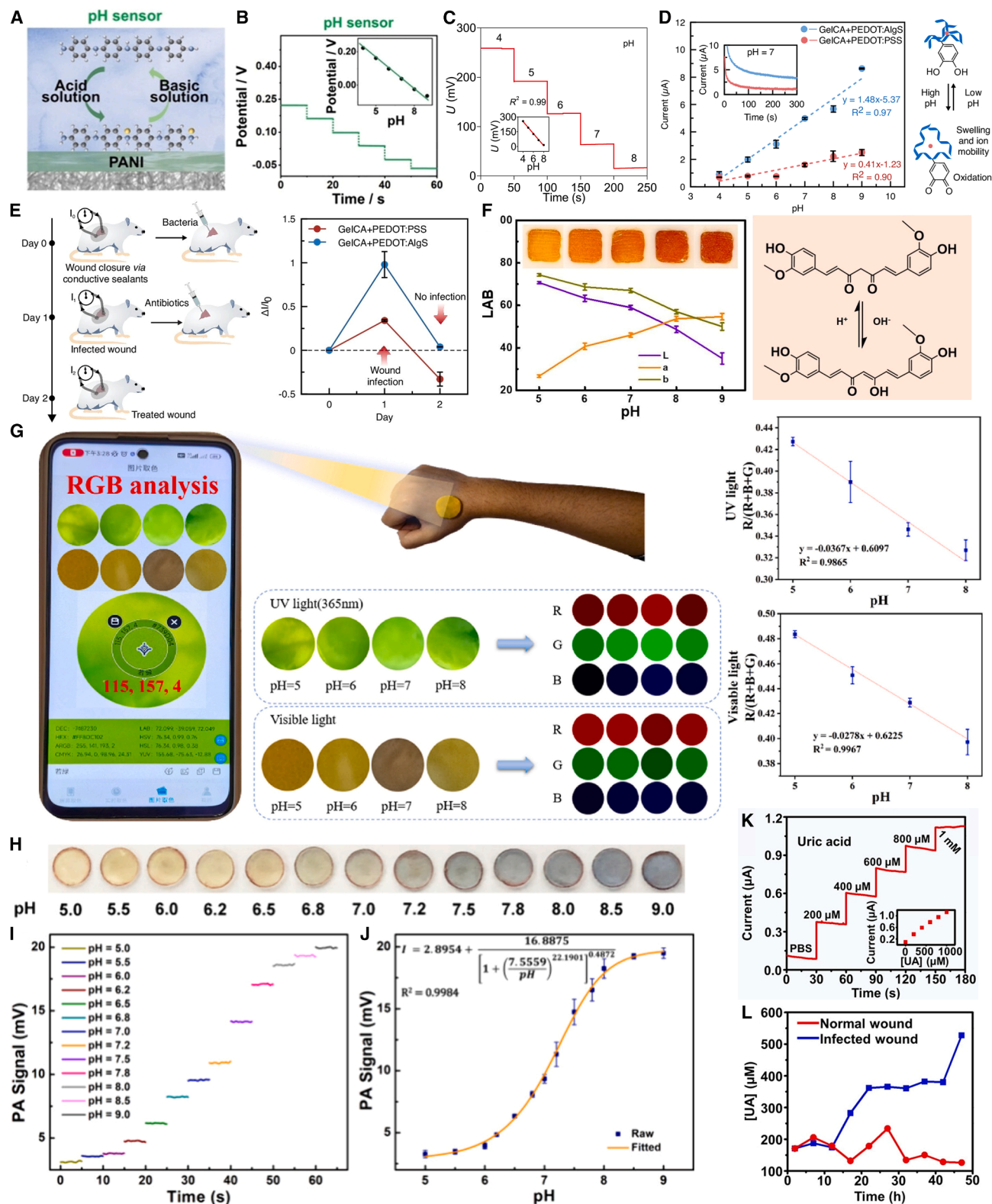
plications. First, long-term stability and mechanical durability need improvement.^{30–34} Second, it is not that easy to maintain a consistent and undisturbed skin-sensor interface, as components of sweat, sebum, shed keratinocytes, and wound exudates can alter interface conditions, leading to measurement inaccuracies.^{14,20} Third, issues related to crosstalk and decoupling among multiple physical parameters remain unresolved. For example, temperature changes may affect impedance and humidity outputs, while pressure changes may alter contact impedance and interfere with temperature or humidity readings.⁵¹ Future research should focus on self-calibrating sensor arrays, surface anti-contamination modifications, and algorithm-based multi-parameter decoupling techniques to achieve long-term, stable, and high-fidelity multimodal sensing.

Small-molecule biochemical signal sensors

pH sensors

In general, normal skin and healed wounds are slightly acidic (pH 4–6),⁶⁴ whereas chronic wounds are typically alkaline (pH 7–9).⁶⁵ Alkaline wounds are prone to bacterial colonization, which can lead to infection and elevate the pH.⁶⁶ Moreover, pH can significantly affects multiple physiological processes during wound healing, including collagen formation, inflammatory response, and angiogenesis.⁶⁷ Existing pH sensors mainly include electrochemical sensors and optical sensors.⁶⁸ Electrochemical sensors generate potential, resistance, or current signals, offering advantages such as low cost, portability, high precision and resolution, and excellent reproducibility.

The pH-responsive materials can convert pH changes into quantifiable electrical signals. Polyaniline (PANI), metal oxides, graphene-based materials, single-walled CNTs (SWCNTs), and poly(3,4-ethylenedioxythiophene) (PEDOT) are commonly used pH-responsive materials that undergo protonation and deprotonation reactions depending on pH variation (Figure 3A).^{14,69} The inherent selectivity of PANI ensures excellent specificity toward protons over other common ions. The pH can be calculated by measuring the potential difference between the working electrode and the reference electrode. For example, Deng et al. prepared a pH sensor by modifying a PANI membrane on screen-printed electrodes. The open circuit potential showed a linear relationship with pH in the range of 4–9 (Figure 3B), matching the physiological pH range of wound exudate.¹⁴ Wang et al. integrated a potentiometric pH sensor based on electropolymerized PANI film, which exhibited a stable potential response and a linear relationship with pH (Figure 3C). pH increased 24 h after bacterial inoculation and then gradually returned to normal after treatment.²⁰ Mariani et al. developed a fully textile-based two-terminal pH sensor, where iridium oxide (IrOx) particles embedded in a PEDOT:poly(styrenesulfonate) (PEDOT:PSS) matrix serve as the sensing layer. Leveraging a potentiometric transduction mechanism driven by spontaneous redox reactions of the $\text{Ir}^{3+}/\text{Ir}^{4+}$ couple and electrochemical gating at the IrOx-PEDOT:PSS interface, the sensor exhibits a sensitivity of $59 \pm 4 \mu\text{A pH}^{-1}$ within the medically relevant pH range of 6–9. It demonstrates excellent robustness against common wound exudate interferents and temperature fluctuations. The referenceless design simplifies integration with flexible textile substrates,



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enabling non-invasive and conformal contact with irregular wound beds.⁷⁰

The pH can also be sensed by measuring conductivity. For example, Chen et al. developed a flexible interdigitated electrode platform based on a laser-ablated graphene/MoS₂ composite whose surface functional groups (e.g., -OH, -COOH) undergo deprotonation at higher pH levels, resulting in an increased density of negative charges on the surface and enhanced charge transfer kinetics. It enables high-performance pH detection over a wide range of pH 2–10 with a sensitivity of approximately 4.7% per pH unit, a response time of around 5.2 s, and excellent long-term stability.⁷⁴ Iversen et al. developed a fully printed, flexible smart patch for wound pH sensing, featuring SWCNTs electrodes printed on a silicon dioxide NP-modified PDMS substrate. This patch eliminates the need for a reference electrode due to the pH-dependent conductivity properties of SWCNTs, achieves a sensitivity of 7.1 Ω/pH in a wound-relevant pH range (5.8–7.8), and exhibits good repeatability and stability. With a mean response time of ~92 s, it enables low-cost and convenient real-time pH monitoring.⁷⁵ Montazerian et al. developed a conductance-based pH sensor by incorporating water-dispersible sulfonated alginate-doped PEDOT into gelatin-catechol hydrogels. This design resolves the poor aqueous dispersibility of conventional PEDOT:PSS, achieving a 5-fold higher dispersion limit and 20-fold conductivity enhancement. It exhibits ~250% greater pH sensitivity (Figure 3D) with a linear response to the physiological pH range and enables an ~3× larger relative current change in response to infection (Figure 3E).¹⁸

Compared to the electrochemical method, optical detection requires no power source or cables, making it relatively straightforward.⁷⁶ The colorimetric method requires the use of pH-responsive dyes. For example, Du et al. utilized the keto-enol tautomerism of the diketone group in curcumin to incorporate pH-responsive curcumin-loaded microbeads into the hydrogel shell of microfiber fabrics. The visible color of the fabric exhibits distinct variations with pH changes, ranging from 5 to 9. (Figure 3F).⁷¹ Additionally, the dual-layer smart dressing developed by Zhu et al. enables real-time visual monitoring by grafting phenol red onto chitosan. The color was transformed into red-

green-blue (RGB) signals and further quantitatively analyzed.⁷⁷ However, there are some disadvantages, including the interference of blood and the difficulties of covalently linking the dyes directly into the matrix to avoid leakage without affecting its pH sensitivity.^{78,79}

Other approaches, such as fluorescent-based or photoacoustic detection, seem to overcome these limitations. Benefitting from the pH sensitivity and self-quenching behavior of fluorescein isothiocyanate (FITC), Giovannini et al. proposed a fluorescence-based optical pH sensor by covalently conjugating FITC onto PDMS polymeric optical fibers to avoid dye leakage. The sensor exhibits a remarkable signal variation to enable clear differentiation between acute and chronic wounds, maintains stable performance under repeated bending, and shows no interference from wound exudate components.⁸⁰ The polyvinyl alcohol (PVA)/sodium alginate/berberine chloride hydrogels designed by Gu et al. display both pH-dependent color/fluorescence responsiveness in visible and ultraviolet (UV) light, enabling real-time pH monitoring (Figure 3G).⁷² However, optical sensors often require frequent recalibration and need additional equipment to obtain accurate values. Guo et al. developed a photoacoustics (PA)-based wearable pH sensor by integrating a pH-responsive PA agent, nitrazine yellow (NY), into a PVA/sucrose (Suc) hydrogel patch. The sensor is capable of detecting pH in the range of 5.0–9.0 with high accuracy after optimization, without interference from blood and wound drugs, and can be integrated with a portable PA system for real-time, on-site chronic wound pH monitoring (Figures 3H–3J).⁷³

Although various pH sensors have been developed, there are still some challenges to be addressed in the future, such as limited sensitivity, poor durability under mechanical stress, complex fabrication processes,^{81–83} deficient biocompatibility,⁸⁴ unsatisfactory operational stability and anti-interference capability, and difficulty in multifunctional integration.⁸⁵

Uric acid sensors

The concentration of uric acid (UA) is closely associated with infection and can reflect the severity of wound infection and metabolism.⁸⁶ In a healthy wound, UA concentration usually ranges between 220 and 750 μM.⁸⁷ Elevated UA levels (>500 μM) indicate a severe wound with excessive reactive

Figure 3. pH-responsive materials

(A) Schematic of a pH sensor based on PANI.

(B) Open circuit potential responses of the pH sensor at different pH levels. Inset: the relevant calibration chart. Reproduced, with permission, from Deng et al.¹⁴ Copyright 2025, American Chemical Society.

(C) Potentiometric response of a PANI-based pH sensor. Inset: calibration plot with a linear fit. From Wang et al.²⁶ Reprinted with permission from AAAS.

(D) *In vitro* pH sensitivity of conductive bioadhesives obtained by chronoamperometric testing of hydrogels at various pH levels. Inset: current variations with time at pH 7 (*n* = 3 independent samples).

(E) *In vivo* monitoring of wound infection using conductive bioadhesive hydrogels. The data represent the mean and SD (*n* = 3 independent samples). Reproduced from Montazerian et al.¹⁸ under a CC BY 4.0 license.

(F) Left: The CIE L*a*b* (LAB) values at pH 5–9 (the insets show the corresponding photographs). Right: The discoloration mechanism of curcumin. Reproduced, with permission, from Du et al.⁷¹ Copyright 2025, Elsevier.

(G) The smartphone-guided real-time pH monitoring function of polyvinyl alcohol/sodium alginate/berberine chloride (PVA/SA/BBR) hydrogels. Reproduced, with permission, from Gu et al.⁷² Copyright 2025, Elsevier.

(H and I) Photographs (H) and PA signals (I) of PVA/Suc/NY hydrogels with different pH values of whole human blood.

(J) Relationship between the PA signal and pH value. Reproduced, with permission, from Guo et al.⁷³ Copyright 2024, American Chemical Society.

(K) Chronoamperometric response of UA sensor in UA solution in PBS.

(L) *In situ* monitoring of UA of the wounds for 48 h after wound creation. Reproduced, with permission, from Liu et al.¹⁹ Copyright 2021, American Chemical Society.

oxygen species (ROS) and inflammation, while decreased UA suggests UA degradation caused by microbial uricase produced by bacterial colonization.⁸⁸ UA sensors need to possess a wide linear range, low limit of detection (LOD), and excellent sensitivity and selectivity. It can be roughly categorized into enzyme-based and enzyme-free detection methods. Enzyme-based sensors are usually developed by immobilizing working electrodes with uricase.⁸⁹ For example, Liu et al. integrated a UA sensor, featuring a Prussian blue (PB)/chitosan mediator layer, and uricase-immobilized working electrode. The sensor operates at a low potential to avoid interference, exhibiting a linear response of 0–1,000 μM , a sensitivity of 1.01 $\mu\text{A}/\text{mM}$ ($R^2 = 0.9891$), and excellent selectivity (Figure 3K). Equipped with real-time temperature compensation and a Nafion protective layer, it maintains stability in complex wound exudate, enabling *in vivo* early infection monitoring by tracking UA concentration spikes in infected wounds (Figure 3L).¹⁹ Enzymatic sensors provide high sensitivity for UA monitoring and excellent selectivity against wound exudate interferents but may increase manufacturing costs and require more stringent operating conditions. Enzyme-free methods can also be employed, such as electrocatalysis, to oxidize UA using carbon-based materials.^{14,25,90}

Reactive species sensors

Key indicators of wound health also include the concentrations of reactive species, such as oxygen (O_2), nitric oxide (NO), and hydrogen peroxide (H_2O_2), which play critical multifaceted roles in inflammation, infection control, angiogenesis, and tissue perfusion.^{91–96} During the initial inflammatory phase, ROS signal to recruit monocytes and macrophages to the wound site, facilitating the removal of debris and pathogens.⁹⁷ NO and H_2O_2 usually reach their peak in the initial stage after injury, coinciding with the influx of inflammatory cells, particularly monocytes and macrophages, into the wound site. However, due to the characteristics of the transient and reactive nature of these species, coupled with the extremely small amount of exudate from the wound, it is difficult to quantify these reactive species.

O_2 sensors. O_2 is an important biomarker for tissue perfusion.^{98,99} Adequate O_2 levels may reduce infection-related biomarker levels, promoting faster wound healing. Moderate reduction in O_2 levels can activate the hypoxia-inducible factor 1 α (HIF-1 α) signaling pathway, promoting angiogenesis and re-epithelialization.⁹⁹ However, chronic or extreme hypoxia is closely associated with exacerbated inflammation, increased infection severity, and impaired healing.¹⁰⁰ Therefore, tissue O_2 saturation can be used as a predictive indicator for surgical wound infection.¹⁰¹ Wound exudate contains dense solids, such as proteins, dead cells, and tissue debris.^{85,102,103} In addition, the complex environment and low O_2 concentration (<10 mmHg) make *in situ* and accurate O_2 detection in raw wound exudate particularly difficult.^{98,104} Electrochemical O_2 sensors are currently the mainstream type of O_2 sensors, but optical sensors that utilize the O_2 -dependent fluorescence/phosphorescence quenching effect of materials can also be used.^{105,106}

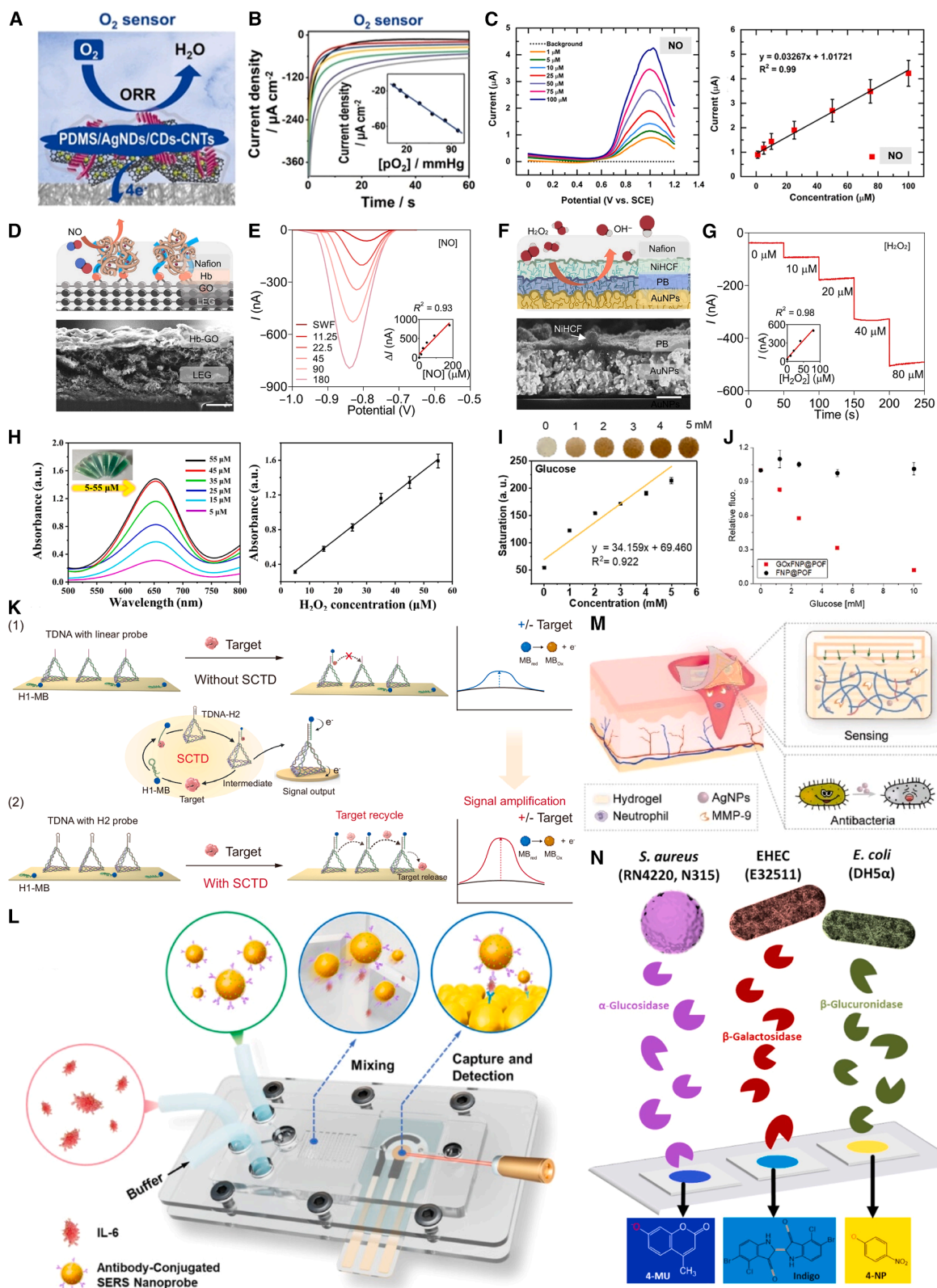
The electrochemical sensing mechanism involves the reduction reaction of O_2 .^{20,107} Deng et al. developed a wound tracker for biomimetic microfluidic exudate sampling as well as noninvasive, skin-conformal, and dynamic monitoring of wound

exudate O_2 .¹⁴ The O_2 sensor adopts a PDMS/silver nanodendrite/carbon dot (CD)-CNT/screen-printed electrode structure (Figure 4A), with PDMS allowing permeation of O_2 but preventing permeation of other biomolecules (such as H_2O_2). The O_2 sensor features a linear detection range of 5–100 mmHg and high selectivity (Figure 4B), achieving stable performance under pH/temperature variations and mechanical deformation. Strong correlations were revealed between O_2 content in wound exudate and UA, lactate, pH, and temperature levels,¹⁴ indicating that hypoxia (1) alters cellular metabolism and increases UA and lactate levels,^{21,108} (2) exacerbates inflammation and raises pH and temperature,^{109,110} and (3) promotes anaerobic bacterium colonization, produces alkaline substances, and increases pH.¹¹⁰

NO sensors. NO, secreted by immunological cells as a defender against bacterial infections¹¹⁴ and can facilitate healing processes.^{115,116} NO acts as a signaling molecule and participates in regulating various physiological processes, such as angiogenesis, virulence, biofilm regulation, and gene expression.^{117,118} In the presence of oxygen, NO converts into reactive nitrogen species, which induce damage to biological macromolecules (e.g., DNA, lipids, and proteins) and control pathogen replication during the later stages of infection.

Most existing NO sensors operate at positive potentials to detect the oxidation of NO. Simoska et al. developed a flexible carbon ultramicroelectrode array (CUA) sensor for electrochemical detection of NO in chronic wounds. Based on square wave voltammetry (SWV) at a characteristic oxidation potential of +1.01 V vs. the reference electrode, the sensor achieves a low LOD of 0.20 μM and a linear dynamic range (LDR) of 1–100 μM (Figure 4C), covering pathophysiological NO concentrations in infected wounds. It enables simultaneous multiplex detection of NO, pyocyanin (PYO; representing bacterial infection), and UA without cross-interference and successfully quantifies NO secreted by macrophages *in vitro*.²⁵ However, the presence of various electroactive interferents in wound exudate, such as H_2O_2 , UA, and amino acids, may significantly reduce the selectivity of NO in this detection mode.^{119–121} To quantitatively measure this transient and highly reactive biomarker in minute volumes of exudate and ensure accurate NO sensing, Wang et al. developed a flexible NO sensor integrated into the iCares, featuring an LEG/graphene oxide (GO) electrode immobilized with hemoglobin (Hb) as the recognition element to ensure high selectivity of NO by specifically catalyzing NO reduction to N_2O , while the LEG enhances electron transfer efficiency (Figure 4D). Operating at a reduction potential of approximately -0.8 V via differential pulse voltammetry (DPV) (Figure 4E), the sensor achieves a low detection limit of 0.5 μM and a linear range of 0–45 μM , covering the physiological NO concentrations in chronic wounds. Integrated with a microfluidic system, the sensor enabled continuous *in situ* NO monitoring in diabetic mouse wound models and 20 human chronic wound patients, among whom 2 patients who underwent vascular intervention showed an increase in NO levels, reflecting improved perfusion.²⁰

H_2O_2 sensors. H_2O_2 is a key biomarker for inflammation and infection. It is also a well-known bactericide, directly killing bacteria through oxidative stress and recruiting immune cells for



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enhanced defense.¹²² Studies have revealed that H₂O₂ can help to kill ingested pathogens and destroy biofilms at an early stage, which is beneficial for wound healing.^{40,123,124} At low concentrations, H₂O₂ facilitates wound angiogenesis *in vivo* by inducing focal adhesion kinase phosphorylation and VEGF production.¹²⁵ However, prolonged or excessive H₂O₂ may exacerbate inflammation and delay healing.

Electrochemical sensors are widely used to detect H₂O₂. PB, displaying high catalytic reduction activity for H₂O₂, is a classic sensitive material for H₂O₂ detection.⁴⁰ The flexible amperometric H₂O₂ sensor integrated in iCares features a nickel hexacyanoferrate/PB-coated AuNP electrode with a Nafion antifouling layer (Figure 4F). The amperometric sensor exhibits a linear correlation with the H₂O₂ concentrations with a sensitivity of 5.91 nA μ M⁻¹ and a low LOD of 0.25 μ M (Figure 4G). Its performance is stable, and H₂O₂ elevation 24 h after bacterial inoculation and recovery after treatment can be monitored.²⁰ Wang et al. developed a flexible H₂O₂ sensor based on Co ion-implanted ZIF-4 MOF film through H₂O₂ oxidation that exhibits a sensitivity of 2,111 μ A mM⁻¹ cm⁻², a LOD of 0.31 μ M, and a linear range of 1–100 μ M with high selectivity against glucose, lactate, and UA.¹²⁶

In addition to electrochemical sensing, Xia et al. developed a flexible colorimetric H₂O₂ sensor based on α -Co@nitrogen-doped carbon nanofiber nanozyme.¹¹¹ The nanozyme exhibits superior peroxidase-like activity, outperforms natural enzyme and other types of nanozymes, catalyzing H₂O₂ to generate $\cdot$ OH radicals that oxidize colorless 3,3',5,5'-tetramethylbenzidine (TMB) into blue oxidized 3,3',5,5'-tetramethylbenzidine (oxTMB). The sensor achieves a linear detection range of 5–55 μ M, a low LOD of 1.66 μ M (Figure 4H), and excellent affinity and catalytic efficiency for TMB and H₂O₂.

Lactate sensors

Lactate is critical for initiating tissue repair processes, such as HIF-1 α stabilization,¹²⁷ growth factor and metalloproteinases production,^{128,129} endothelial cell mobility,¹³⁰ vascularization,¹²⁷ increased collagen synthesis,^{129,131} and others.^{132,133} The concentration of lactate in normal tissues is not yet fully clear. Some studies suggest it to be 1–3 mM, while others indicate it to be between 5 and 15 mM. A study has shown that fibroblasts may be impaired when lactate increases above 7 mM.¹³⁴ Hypoxia can lead to an increase in lactate levels, delaying wound healing, while low lactate level may impede initiation of healing process.¹²⁷

Enzyme-based bio-specific recognition sensors are currently the mainstream type for monitoring lactate levels in wounds. Garland et al. developed a lactate sensor immobilizing tetrathiafulvalene and lactate oxidase on the CNT paper as the anode and oxygen-independent silver oxide as the cathode based on a self-powered biofuel cell, achieving an LOD of 0.47 mM and negligible interference from glucose, UA, and ascorbic acid. A developed machine learning model trained on the lactate data can predict wound closure rate with $\sim$ 76% accuracy. The accuracy can be improved to $\sim$ 83% if combined with additional training on pH.²¹ However, the activity of lactate oxidase is susceptible to fluctuations, such as temperature and pH, and insufficient oxygen may lead to errors.^{14,107} Therefore, real-time calibration may be necessary. Alternatively, a lactate dehydrogenase-based sensor can be used as an alternative,¹³⁵ but it still has some drawbacks, including reagent instability and high reversibility of the reaction lactate-pyruvate. The use of engineered enzyme may hold the potential to overcome these issues and outperform traditional enzyme-based sensors in sensitivity and stability.¹³⁶

Figure 4. Reactive species sensors

(A) Schematic of an O₂ sensor (PMDS/AgNDs/CDs-CNTs/SPE) that is based on the O₂ reduction reaction principle.

(B) Current responses of the O₂ sensor at different O₂ levels. Inset: the relevant calibration chart. Reproduced, with permission, from Deng et al.¹⁴ Copyright 2025, American Chemical Society.

(C) Left: SWV current responses (background subtracted) for biologically relevant concentrations of NO in simulated wound media. The peak at +1.01 V vs. SCE corresponds to the NO redox reaction. The corresponding calibration curves for NO. Right: The error bars represent the SD between six samples. Reproduced, with permission, from Simoska et al.²⁵ Copyright 2020, American Chemical Society.

(D) Schematic of the mechanism and scanning electron microscopy (SEM) images of an LEG/GO electrode for NO sensing. Hb, hemoglobin. The thickness of the Hb-GO layer is 8 μ m, and that of the LEG layer is 25 μ m. Scale bar, 15 μ m.

(E) Representative DPV responses of the NO sensor in simulated wound fluid. Inset: calibration plot with a linear fit. I, current; Δ I, peak current density height.

(F) Schematic and SEM image of a NiHCF/PB-coated AuNP electrode for H₂O₂ detection. The thickness of the PB layer is 30 nm, and that of the AuNP layer is 600 nm. Scale bar, 400 nm.

(G) Amperometric responses of the H₂O₂ sensor in SWF. Inset: calibration plot with a linear fit. From Wang et al.²⁶ Reprinted with permission from AAAS.

(H) UV-visible (UV-vis) absorbance spectra and corresponding calibration curves for H₂O₂. Reproduced, with permission, from Xia et al.¹¹¹ Copyright 2024, American Chemical Society.

(I) Calibrations of glucose. The error bars represent the SD for three determinations. The output signals saturation values for glucose. Reproduced, with permission, from Zheng et al.²² Copyright 2023, American Chemical Society.

(J) Confirmation of the efficiency of GOx/FNP@POF in detecting glucose. Reproduced, with permission, from Giovannini et al.⁸⁰ Copyright 2023, Wiley.

(K) Schematic comparison of DNA circuit reactions and sensing signals triggered by target with (2) and without (1) SCTD, along with an illustration of the SCTD working principle. In hydrophilic confined regions, target proteins in the wound exudate trigger DNA cascade amplification around electrode-anchored transfer DNA, with pre-coated auxiliary DNA (H1) dissolution enabling the amplification process. Reproduced from Zhao et al.¹¹² under a CC BY-NC-NC 4.0 license.

(L) Conceptual illustration of the dual-mode microfluidics platform with distinct structural functional areas for detecting IL-6 in diabetic wound exudates. Reproduced, with permission, from Chen et al.²³ Copyright 2025, American Chemical Society.

(M) Schematic of the smart wound dressing for wearable matrix metalloproteinase-9 (MMP-9) sensing and controlled release of the antibacterial drug. Reproduced, with permission, from Deng et al.²⁴ Copyright 2025, Elsevier.

(N) The release of the dyes, which are characterized by different colors, is caused by cleavage reactions catalyzed by the corresponding target enzymes secreted from *S. aureus* RN4220, EHEC E32511 methicillin-resistant *S. aureus* (MRSA) N315, enterohemorrhagic *aureus* (MRSA) N315, enterohemorrhagic *E. coli* (EHEC) E32511 (EHEC) E32511, and *E. coli* DH5 α , respectively. Reproduced, with permission, from Jia et al.¹¹³ Copyright 2021, Elsevier.

Glucose sensors

Diabetic wounds often suffer from impaired healing due to hyperglycemia. Elevated blood glucose levels delay wound healing through multiple pathological mechanisms, including pro-inflammatory states, tissue hypoxia, osmotic diuresis, impaired vascular function, and peripheral neuropathy.¹³⁷ Since glucose concentration in wound exudate correlates strongly with blood glucose levels, real-time and accurate monitoring of glucose in wound exudate can provide direct and dynamic guidance for clinical intervention. Based on whether enzyme catalysis is employed, glucose sensors can be classified into enzyme-based and non-enzyme-based types.¹³⁸

Non-enzymatic sensors typically utilize the electrocatalytic activity of metal, metal oxide, alloy metals, and nanomaterials to directly catalyze glucose. For example, Wang et al. developed a glucose sensor based on an MOF-derived 3D hollow composite. The sensor exhibits two linear detection ranges (0.5–722 and 722–4222 μM) with corresponding sensitivities of 4.05 and 1.43 $\mu\text{A L } \mu\text{mol}^{-1} \text{ cm}^{-2}$ and a low LOD of 0.83 μM . Benefitting from the synergistic electrocatalysis of FeOOH and copper-cobalt-layered double hydroxide, as well as the hollow structure's abundant active sites, it shows excellent anti-interference against UA, lactate, and metal ions.¹³⁹ Although non-enzymatic sensors exhibit superior stability, their selectivity is relatively poor when compared with enzyme-based sensors, being susceptible to interference from other biomarkers.¹⁴⁰

Currently, enzymatic sensors rely on the specific catalytic activity of biological enzymes. Glucose oxidase (GOx) oxidizes glucose and generates measurable electrometric or optical signals. The combination of GOx as the specific recognition element and PB as the electron mediator is commonly used for glucose detection.^{141,142} Moreover, the organic electrochemical transistor transducer with high transconductance holds the potential to enhance the signal-to-noise ratio for reliable detection.¹⁴³

An enzymatic reaction in combination with a redox-sensitive chromogenic substrate can be used to realize colorimetric detection. Zheng et al. developed a colorimetric glucose sensor constructed by encapsulating GOx and horseradish peroxidase within an asparagine-poly-L-lysine CD-doped agarose hydrogel, which enhances enzyme stability and maintains substrate affinity. It exhibits a linear detection range of 0–5 mM with an LOD of 0.27 mM, and the color changes from pale yellow to dark yellowish brown with increasing glucose concentration (Figure 4I), quantifiable via saturation analysis.²² A fluorescence-based strategy can also be employed. A wearable glucose sensor is constructed by covalently functionalizing PDMS polymer optical fibers with GOx-modified FITC-loaded silica NPs. The detection mechanism relies on GOx-catalyzed oxidation of glucose to gluconic acid, which reduces the fluorescence intensity of pH-sensitive FITC in a concentration-dependent manner. It exhibits a linear detection range of 0.5–10 mM, an LOD of 0.17 mM (Figure 4J), and high selectivity for glucose.^{80,144}

Volatile organic compound sensors

Wounds and their surface microbiomes continuously release various volatile organic compounds (VOCs), such as ammonia, hydrogen sulfide, and acetic acid, which mainly come from the

decomposition of bacteria on carbohydrates, fats, proteins, amino acids, and other substances. Since each bacterial species emits a distinct VOCs profile, analyzing VOCs at the wound site can not only indicate infection but also specifically identify the infection type, enabling early and timely intervention. For example, although methicillin-sensitive *Staphylococcus aureus* (MSSA) and *Pseudomonas aeruginosa* both produce VOCs containing 2-methyl-1-propanol and 3-methyl-1-butanol, 3-methylbutanal is unique to MSSA. The VOCs of *P. aeruginosa* include hydrogen cyanide, 5-methyl-2-hexanamine, and 2-nonanone, enabling differentiation from MSSA. The unique VOC profile of each bacterium holds significant potential for early wound infection identification, facilitating subsequent targeted antibiotic therapy.¹⁴⁵ Sun et al. developed a collaborative wound infection detection system integrating an electronic nose and field asymmetric ion mobility spectrometry (FAIMS). The system can directly detect characteristic VOCs and many other sensitive chemicals produced by three common wound pathogens (*Escherichia coli*, *P. aeruginosa*, and *S. aureus*), using a 35-sensor array for broad odor profiling and FAIMS for ultra-sensitive parts-per-trillion (ppt)-level specific ion separation. A dedicated information fusion algorithm integrates heterogeneous data, achieving excellent accuracy, sensitivity, and specificity 24 and 48 h post inoculation.¹⁴⁶

To address the limitations of conventional wound monitoring, which relies on physical contact and fails to capture VOC-related signals, Shin et al. developed a non-contact wearable epidermal flux sensor for real-time VOC flux monitoring. The device features an enclosed microclimate chamber adjacent to the skin, a programmable bistable valve, and integrated wireless sensors. By dynamically modulating the chamber's connection to the environment, it quantifies VOCs flux density via the initial concentration change rate after valve closure, avoiding physical coupling with the wound. Validated in wound models, including healthy, diabetic, and infected mice, and human trials, the sensor detects characteristic VOC flux variations: a transient increase during the inflammatory phase of healthy wound healing, anomalous flux patterns in diabetic wounds with dysregulated inflammation, as well as an ~ 100 -fold flux elevation in *S. aureus*-infected wounds and reversion after antibacterial intervention. This non-invasive design enables safe, continuous VOC monitoring, providing critical insights into wound inflammation, infection, and healing progression without disrupting the wound bed.¹⁴⁷

Bottlenecks

The practical application of small-molecule biosensors faces several bottlenecks. First, preventing biofouling is a major challenge; wound exudate, which is rich in proteins, lipids, cellular debris, and microorganisms, can adsorb onto the electrodes, reducing the effective surface area, hindering electron transfer, and leading to signal attenuation or failure. Although anti-fouling coatings have been explored, their stability and versatility in dynamic wound environments, as well as their impact on sensitivity and response speed, remain unclear.^{19,20} Second, enzyme-based sensors struggle to maintain long-term activity because oxidases and dehydrogenases are vulnerable to lose their activity. Advancements in enzyme immobilization techniques and the development of stable artificial enzymes are crucial for practical

implementation.^{19–21,90,119–121} Third, issues of cross-interference and signal crosstalk in multiparameter detection remain unresolved. Electroactive substances in exudates have overlapping redox potentials, making it difficult to distinguish them using single-potential detection. Selective membranes, specific enzymes, or multimodal signals can mitigate this issue, but they increase complexity and cost.^{20,25} Future development should focus on novel sensing materials with high selectivity, anti-fouling capability, and long-term stability and integration with microfluidic chips for sample pretreatment and separation to enhance detection reliability.

Macromolecule sensors

Protein sensors

Cells engaged in the healing process release several proteins, such as cytokines, including tumor necrosis factor alpha (TNF- α) and interleukin (IL), and enzymes, including matrix metalloproteinase (MMP), elastase, and bacterium-specific proteases. Protein biomarkers, dynamically reflecting the progression of inflammation, tissue remodeling, infection status, and cellular metabolism, can serve as important indicators of wound conditions.¹⁴⁸

Currently, the specific molecular recognition mechanism based on aptamers and antibodies is often used for protein detection. Gao et al. developed a multiplexed sensor array targeting multiple protein biomarkers, including TNF- α , IL-6, IL-8, and transforming growth factor β 1 (TGF- β 1), utilizing specific aptamers modified with methylene blue (MB) as redox probes. The sensing mechanism relies on aptamer conformational changes upon target binding, which alter the distance between MB and the graphene-gold NP-modified electrode, leading to measurable changes in redox current. High-fidelity wound biosensing is challenging because of the current biosensors' limitations in detecting low-abundance proteins. To solve this problem, Zhao et al. developed a protein sensor integrated with a self-confined tetrahedral DNA circuit, with hydrophilic regions confining auxiliary hairpin DNA (containing a target-specific recognition aptamer sequence) and tetrahedral DNA anchored on gold electrodes. Target proteins can trigger cyclic amplification of the DNA circuit, realizing recyclable utilization of targets and enhancing electrochemical signals from MB (Figure 4K). It simultaneously detects TNF- α , IL-6, TGF- β 1, and vascular endothelial growth factor with a low LOD and high correlation coefficient ($R^2 > 0.98$).¹¹² In addition, in order to address interference from miscellaneous proteins and insufficient accuracy in quantifying low-concentration IL-6, Chen et al. developed a neural network-enhanced dual-mode microfluidic sensor for ultrasensitive detection of IL-6 based on antigen-antibody binding. The sensor integrates electrochemical and surface-enhanced Raman spectroscopy (SERS) techniques, with antibody-conjugated nanoprobe forming sandwich immunocomplexes with IL-6 on the working electrode. A square-wave micromixer with nonleaky obstacles enhances the binding efficiency between IL-6 and the nanoprobe (Figure 4L). It achieves a detection limit of 0.085 pg/mL for the electrochemical mode and 0.047 pg/mL for the SERS mode, with a linear detection range of 0.05–1,000 pg/mL. A self-attention mechanism-equipped neural network dynamically assigns weights to data from both modes, resulting in a quantitative accuracy of up to 99.8%.²³

In addition to biological recognition elements, synthetic recognition elements can also be used. Lourenço et al. developed an electrochemical biosensor integrated with molecularly imprinted polymers to monitor IL-6. The polymer-based sensing layer, fabricated via electropolymerization of pyrrole and pyrrole-3-carboxylic acid to enhance electrostatic interactions with IL-6, exhibits a linear response in the range of 50 pg mL⁻¹ to 50 ng mL⁻¹ and a low LOD of 8 pg mL⁻¹ along with high selectivity against interferents like C-reactive protein and glucose.¹⁴⁹

Enzyme sensors

Enzymes can degrade substrates to generate signals. MMPs are kinds of protein biomarkers related to cellular metabolism. During normal healing, MMP expression is tightly regulated to promote the clearance of necrotic tissue and formation of new tissue. However, in chronic wounds, the activity of MMPs (particularly MMP-2 and MMP-9) is usually elevated, leading to excessive extracellular matrix degradation, severely impeding the healing process.¹⁵⁰ Deng et al. developed a radiofrequency MMP-9 sensor based on a bioresponsive hydrogel composed of methacrylic anhydride-modified gelatin, sodium alginate, and silver NPs, which can be specifically degraded by MMP-9. The degradation of the hydrogel modulates the capacitance of the interdigitated electrode (Figure 4M). It exhibits a wide linear detection range of 0.01–1 μ g/mL and 3–25 μ g/mL, a low LOD of 8.9 ng/mL, and excellent selectivity against interfering substances. Additionally, the hydrogel enables controlled release of silver NPs upon degradation, exhibiting significant antibacterial activity against *E. coli* and *S. aureus*.²⁴ Giovannini et al. developed an MMP sensor fabricated by functionalizing PDMS polymer optical fibers with FITC-loaded fluorescent gelatin-chitosan NPs. It exhibits a linear detection range of 5.4–10 μ g/mL and an LOD of 1.8 μ g/mL. The detection mechanism relies on MMP-mediated enzymatic degradation of the gelatin matrix, which reduces the self-quenching effect of FITC and enhances fluorescence intensity.^{72,144}

Virulence factor sensors

P. aeruginosa is one of the most commonly microorganisms among hospital-acquired infection, particularly in patients suffering from chronic wounds.^{151–155} During early infection stages, *P. aeruginosa* produces various redox-active phenazine derivatives as virulence factors,^{156–159} among which PYO is a specific molecule marker.^{156,160} Due to its unique redox characteristics, PYO readily reacts with cellular metabolites to generate ROS, causing damage to host organisms and tissue.^{156,161,162} The flexible CUAs developed by Simoska et al. were applied to identify and monitor PYO electrochemically in simulated wound medium and cell cultures, with a LOD of 1.0 μ M and LDR of 1–250 μ M, enabling rapid diagnosis of *P. aeruginosa* infection. A significant decrease of PYO was detected by CUAs after being treated with silver (Ag⁺).²⁵

Additionally, bacterial metabolic processes release characteristic enzymes, and identification of these specific biomarkers enables accurate assessment of bacterial colonization and wound infection. Opportunistic pathogens, such as *S. aureus*, *P. aeruginosa*, and *Streptococcus pyogenes*, are commonly implicated in wound infection and can secrete a virulence factor called deoxyribonuclease (DNase). To detect wound infection by targeting DNase, Xiong et al. developed a DNase sensor

Table 2. Representative studies of flexible bioelectronic systems for treatment

Representative study	Treatment strategy	Validation level	Key advantages	Key limitations	Translational potential
Ge et al. ¹⁶⁶	stimulus-responsive drug delivery	<i>in vitro</i> , animal model	on-demand release, effective against <i>S. aureus</i> and <i>E. coli</i>	requires external voltage for heating; temperature control precision needs optimization	medium; it can be suitable for chronic wounds
Su et al. ¹⁶⁴	stimulus-responsive drug delivery	<i>in vitro</i> , animal model	voltage-controlled, rapid release, and good antibacterial effect	drug loading capacity is limited; long-term safety needs validation	high; it can be suitable for infected wounds
Jiang et al. ¹⁶	electrical stimulation	<i>in vitro</i> , animal model	increases healing speed by approximately 25% and promotes the expression of regenerative genes	requires 6 h daily stimulation, compliance to be improved	high; it can be integrated into wireless closed-loop system
Kaveti et al. ¹⁶⁵	electrical stimulation	<i>in vitro</i> , animal model	water-activated, no external electronics	single-use, electric field intensity not adjustable	high; it can be suitable for home care
Kang et al. ¹⁵	thermotherapy	<i>in vitro</i> , animal model	rapid heating to promote vasodilation (reaches 40°C in 20 s)	temperature control accuracy needs improvement	medium; it requires temperature feedback integration
Lyu et al. ¹⁶⁶	ultrasound treatment	<i>in vitro</i> , animal model (type II diabetes)	non-invasive, deep penetration, activates the Rac1 signaling pathway	the device is complex and requires external power	medium; it needs to be further miniaturized
Xue et al. ¹⁶⁷	light therapy	<i>in vitro</i> , animal model (infected wound)	drug-free, promotes angiogenesis and collagen deposition	requires external power, limited light penetration depth	medium; it can be suitable for superficial infections

based on a chemically crosslinked DNA hydrogel that degrades upon exposure to DNase, leading to changes in dielectric permittivity and capacitance of the interdigitated electrode. The sensor can detect clinically relevant amounts of *S. aureus* both *in vitro* and *in vivo* before visible manifestation of infection.¹⁶³

In addition, the colorimetric approach by incorporating colorimetric enzyme substrates enables bacterial enzyme sensing visually. Yu et al. developed a sensor hydrogel functionalized with three different colorimetric enzyme substrates, which can be degraded by α -glucosidase, β -galactosidase, and β -glucuronidase secreted from bacteria, effectively detecting and differentiating different bacterial strains and infection levels with high sensitivity and selectivity through significant color changes (Figure 4N). This hydrogel can simultaneously detect all three characteristic enzymes within 60 min.¹¹³

Bottlenecks

In general, sensors for macromolecular biomarkers face greater challenges than those for small molecules. First, the stability and reusability of the recognition elements are limited.^{23,112,149} Second, the sensitivity of detecting low-abundance proteins remains challenging. The key cytokines in wound exudate (e.g., IL-6 and TNF- α) are present at extremely low concentrations, and the LOD of conventional electrochemical or optical sensors often fails to meet the requirements of early clinical diagnosis.^{23,112} Third, system integration and anti-interference capability for

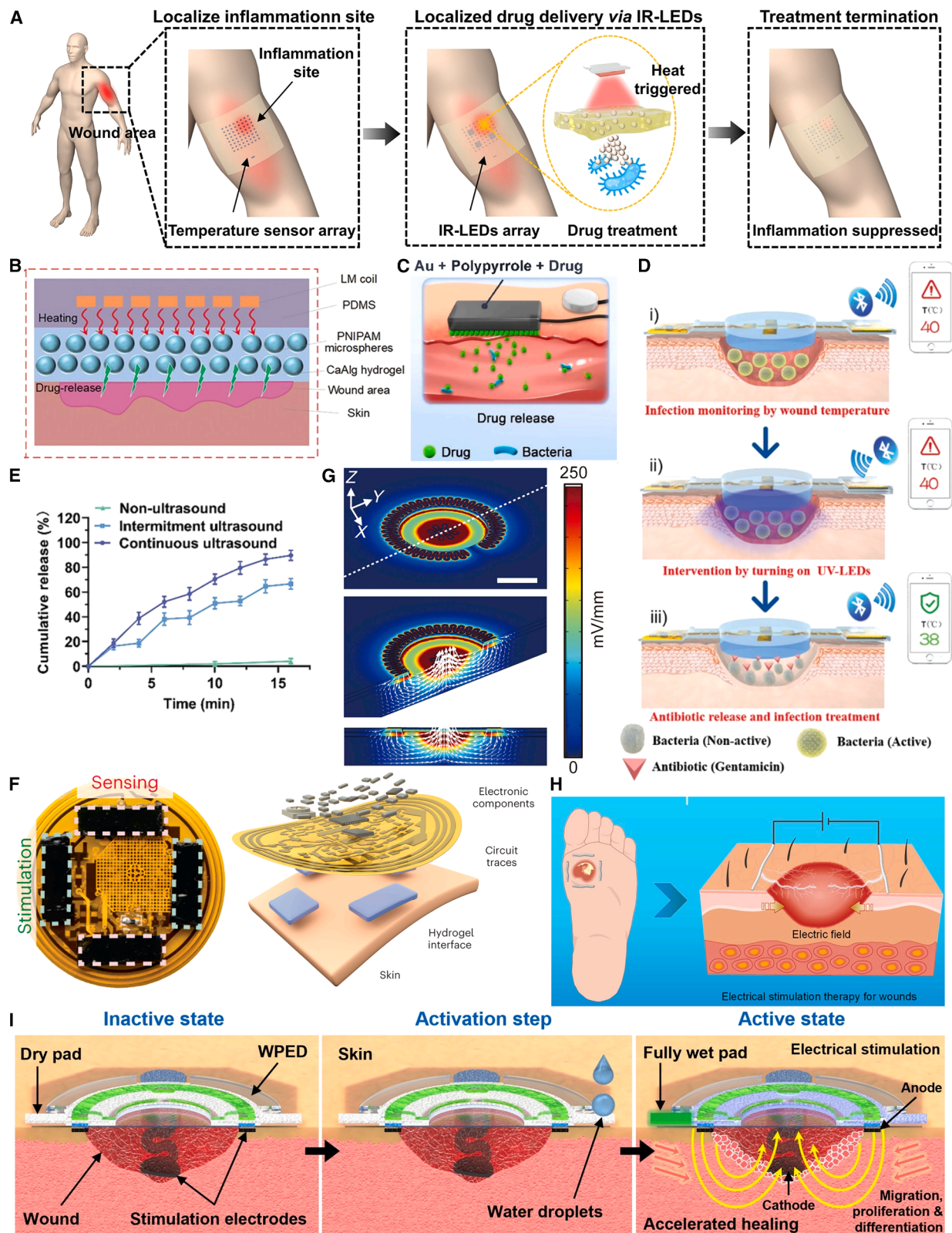
simultaneous multi-parameter detection are insufficient. Multiplexing macromolecule sensors typically requires multiple working electrodes or distinct labeled probes, which not only increases system complexity but also exacerbates the risks of cross reactions and non-specific adsorption.^{85,112} Future research should prioritize the development of stable, high-affinity synthetic recognition elements, microfluidic sample processing technologies, and orthogonal signals to ensure accuracy, although miniaturization and cost remain key barriers in translation.²³

TREATMENT

Flexible bioelectronic systems enable multiple therapeutic modalities for wound care, including electrical stimulation, thermotherapy, ultrasound treatment, and stimuli-responsive drug delivery, with representative studies summarized in Table 2.

Stimulus-responsive drug delivery systems

Stimulus-responsive drug delivery systems (DDSs) are mainly composed of hydrogels or nanomaterials, enabling controlled drug release upon specific stimuli, which are generally categorized into endogenous and exogenous stimuli. Thermosensitive biomaterials, such as poly(N-isopropyl acrylamide) (PNIPAM) and Pluronic, with the property of low critical solution temperature have been widely used for controlled drug delivery.



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Endogenous stimuli are derived from the wound microenvironment, including temperature (increased in infected/inflamed wounds),¹⁶⁸ pH (low in infected/inflamed wounds),^{169,170} ROS (high in chronic nonhealing wounds),¹⁶⁹ enzymes (e.g., MMPs),¹⁷¹ and glucose (elevated in diabetic wounds).¹⁷⁰ The most used endogenous stimuli are pH and temperature.¹⁷²

Here, we introduced some external stimuli, including heat, electricity, light, ultrasound, and magnetic field. A heat-triggered drug delivery module consists of an amoxicillin-loaded gelatin hydrogel and an infrared light-emitting diode (IR-LED) array. After localizing the inflammation regions associated with temperature change, the IR-LED can generate localized heat to induce solution-to-gel (sol-gel) transition of the hydrogel, enabling targeted antibiotic release (Figure 5A). Suppressed *S. aureus*-induced infection, higher collagen volume, and reduced inflammatory cell infiltration were confirmed *in vivo*.²⁷ Thermosensitive and cefazolin-loaded PNIPAM-co-acrylamide hydrogel microspheres embedded in a hydrogel patch have been developed to achieve controlled drug release (Figure 5B). After being heated by the liquid metal coil via voltage, the microspheres can shrink and release antibiotics, achieving on-demand drug release and effectively inhibiting *S. aureus* and *E. coli*.²⁶

Su et al. developed an electrically controlled antibiotic release system using polypyrrole as the drug carrier. During electropolymerization, cefazolin sodium was incorporated into the polypyrrole matrix on the gold particle-based electrodes (Figure 5C). The drug release is precisely controlled by voltage. When a -0.5 V voltage is applied to the electrode, the redox reaction of polypyrrole disrupts the interaction between the polymer matrix and the drug molecules, triggering rapid release of cefazolin sodium. The smart dressing can effectively inhibit *S. aureus* proliferation, reduce inflammatory cell infiltration, and promote wound healing.¹⁶⁴

A UV-responsive antibacterial hydrogel is integrated in the lower layer of the system for on-demand release. Gentamicin was conjugated with a polyethylene glycol (PEG)-based hydrogel by a UV-cleavable nitrobenzyl linker to obtain a polyprodrug (gentamicin-linker-PEG-acrylate), which was subsequently crosslinked by acrylate-PEG acrylate (PEGDA) to form the hy-

drogel, enabling gentamicin release upon 365 nm UV irradiation from the UV-LEDs in the upper layer triggered by infection-induced temperature elevation (Figure 5D). Reduced densities of *S. aureus* colonies and inflammatory cells were verified *in vivo*.¹⁷³

Ultrasound-triggered drug delivery has emerged as a new versatile strategy for on-demand wound therapy. Huang et al. designed injectable hydrogels integrating poly(L-lactic acid) microcapsules loaded with cefazolin sodium. These microcapsules undergo vibration-induced rupture under ultrasonic irradiation, enabling spatiotemporally controlled drug release, with $\sim 90\%$ release within 16 min under continuous ultrasound and $<5\%$ release without stimulation (Figure 5E). *In vivo* studies of infected and diabetic wounds confirm enhanced healing, reduced inflammatory cytokines, and promoted collagen deposition.¹⁷⁴

The inherent differences among these stimuli lie in their sources (which are endogenous and exogenous), control precision, and response speed. Endogenous stimuli can activate drug release without external devices and are site specific, making them suitable for various types of wounds. Although the design of DDSs has improved, the drug release rate from endogenous stimulus-responsive materials is predetermined by the encapsulation matrix, irrespective of patient needs or changing physiological circumstances.¹⁷⁶ Furthermore, the physiological conditions may not change significantly to enable quick and efficient drug release at the early stage of infection. For example, the environment and individual temperature variations may affect the delivery precision of temperature-responsive materials. Compared with endogenous DDSs, exogenous DDSs may be more stable and more likely to demonstrate comparable efficacy under clinical conditions because they are able to more precisely control the timing, location, and dosage of drug release. For example, electricity features rapid response and excellent integration with flexible electronics, facilitating the construction of closed-loop systems and making it easier to translate. Moreover, electricity can also be used as a tool to generate other stimuli, and ES can even work in synergy with drug release to promote wound healing. Therefore, electricity is probably the most widely adopted among so many types of exogenous stimuli. Light and

Figure 5. Treatment

(A) Schematic of the operation of inflammation monitoring and treatment (IMT) systems. The temperature sensor array detects a local temperature increase in the inflamed area, activating the IR-LED for *in situ* heat-triggered drug release from the hydrogel until inflammation is suppressed. Reproduced, with permission, from Liu et al.²⁷ Copyright 2024, National Academy of Sciences.

(B) Schematic illustrating encapsulation of drug-loaded thermosensitive hydrogel microspheres in Ca-Alg hydrogel. Reproduced, with permission, from Ge et al.²⁶ Copyright 2023, Wiley.

(C) Schematic of the smart dressing for providing treatment via electrically controlled drug release. Reproduced, with permission, from Su et al.¹⁶⁴ Copyright 2024, Elsevier.

(D) Conceptual view of the integrated system for infected wound monitoring and on-demand treatment: i, real-time monitoring of wound temperature and an alert of hyperthermia caused by infection; ii, turning on UV-LEDs to trigger the release of antibiotics; iii, infection inhibition by the released antibiotics, resulting in decreased wound temperature. Reproduced, with permission, from Pang et al.¹⁷³ Copyright 2020, Wiley.

(E) *In vitro* release profiles of cefazolin from the hydrogel at different conditions with continuous and intermittent ultrasound at a power of 200 W. Reproduced, with permission, from Huang et al.¹⁷⁴ Copyright 2024, Wiley.

(F) Left, photographs of the back side of the smart bandage, showing the stimulation and sensing electrodes. Right, exploded view of the wireless smart bandage, including FPCB and tissue-interfacing conducting adhesive hydrogel. Reproduced, with permission, from Jiang et al.¹⁶ Copyright 2023, Springer Nature.

(G) Finite element analysis results of the electric field between the positive (+) and negative (−) electrodes. Scale bar, 3 mm. Reproduced from Song et al.^{29,175} under a CC BY 4.0 license.

(H) Fiber electrodes applied to a foot wound. I, schematic of wound healing accelerated by an electric field. Reproduced, with permission, from Ge et al.⁴⁷ Copyright 2025, Wiley.

(I) Cross-sectional view illustrations showing the activation and working mechanism of WPEDs. Reproduced from Kaveti et al.¹⁶⁵ under a CC BY-NC 4.0 license.

ultrasound can penetrate deeper layers. Magnetic stimulation also enables noninvasive deep control but requires specific equipment. Whether endogenous stimulus or exogenous stimulus, it should be noted that long-term exposure to wound micro-environments containing various active chemicals may affect the stability and therapeutic activity of the loaded drugs.

Although progress has been made, there are still some issues that need to be addressed about DDSs. First, the dosage of most DDSs is not unlimited, limiting their long-term use. Since most DDSs release drugs through material degradation, once the hydrogel completely degrades, the DDS will lose its ability to completely release drugs as well. Therefore, it is necessary to further develop reusable, stable, and controllable DDSs.¹⁷⁷ Second, the “on-off” or pulsatile release profile may not be able to maintain the drug concentration at a long-term stable level, which may lead to inconsistent therapeutic efficacy or drug waste.^{26,27,164} Therefore, it is necessary to improve the precision of release kinetics and the ability to achieve sustained, programmable release. Third, the biocompatibility and degradation product safety of drug carriers require long-term validation.^{26,164,174} In addition to antibiotics, there are also various biologically active substances that can be used to modulate wound healing, such as small interfering RNA (siRNA)¹⁷⁸ and growth factors.^{171,179} However, they are more fragile compared with drugs. Issues such as susceptibility to degradation, inactivation, and short half-life need to be addressed. Future development should focus on intelligent delivery systems with closed-loop control, integrating real-time sensor data with release control algorithms to achieve dynamic dose adjustment according to wound status.

ES

After skin injury, an endogenous electrical field is generated to initiate and promote wound healing, but it tends to weaken or become disordered in chronic wounds. ES can enhance epidermal thickness, promote angiogenesis, increase collagen deposition, modulate cell behavior and inflammation state, and combat infection to accelerate tissue repair,^{16,29,47,165,180–186} providing a safe and effective solution for treating chronic wounds. For example, a rectified alternating current (AC) signal oscillating between 0 and 2 V was delivered on wounds via hydrogel electrodes (Figure 5F). It was observed that it accelerated wound closure by ~25%, enhanced dermal remodeling by ~50%, and increased neovascularization and proregenerative gene (CD163 and MRC1) activation and a shift to the anti-inflammatory M2 phenotype in monocytes/macrophages after daily 6-h ES.¹⁶ In addition, direct current (DC) can also be used. The electrotherapy system that can deliver 1.1 V DC ES can accelerate diabetic wound closure by ~30%, enhance granulation tissue thickness, and promote neovascularization through daily 30-min stimulation (Figure 5G).²⁹ Similarly, the outer liquid metal (LM)-copper alloy coating of the dual-layer fiber serves as electrodes to deliver 0.6–5 V DC for 30 min/day, with a typical 2 V output (approximately 200 mV/mm electric field), mimicking endogenous bioelectric fields (Figure 5H).⁴⁷ Kaveti et al. developed a water-powered, electronics-free dressing (WPED) to generate a radial DC electric field (peak ~530 mV/mm near the wound center, ≥100 mV/mm at the periphery) through stimulation electrodes (Figure 5I).¹⁶⁵

Although ES holds great promise as a non-invasive, drug-free physical therapy for wound healing, it still faces several technical challenges. First, there is a lack of standardization regarding optimal stimulation parameters; variations in stimulation type, intensity, frequency, and duration across different studies, as well as the absence of parameter optimization for different wound types, hinder the development of a unified clinical protocol.^{16,29,47} Second, the interface between the electrodes and the wound requires greater stability and comfort, as existing materials (such as hydrogels and metal films) suffer from issues like reduced adhesion, ion migration, and mechanical fatigue in the moist wound environment.^{16,29,47}

Thermotherapy

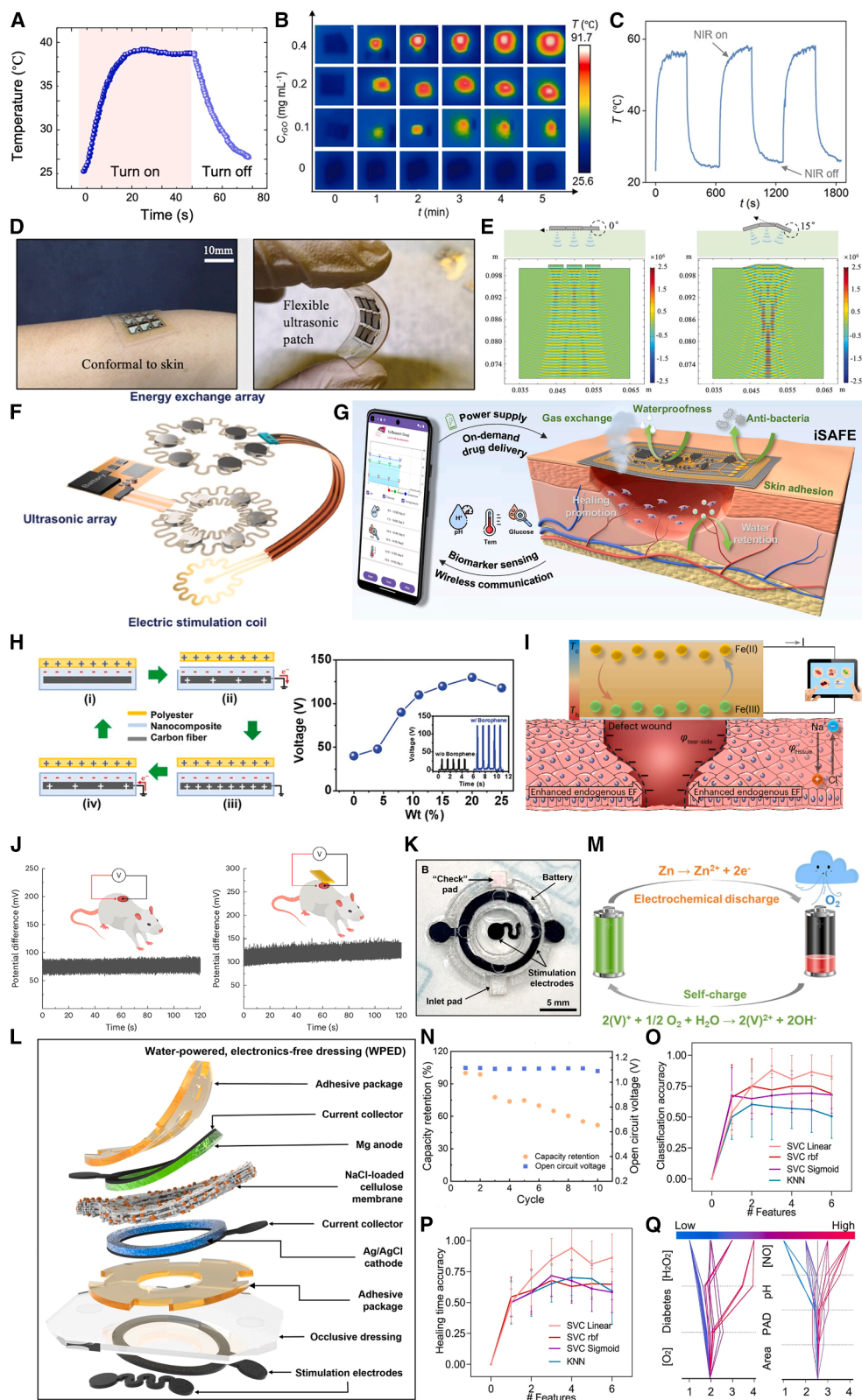
Thermal therapy mainly includes direct heating, photothermal treatment, etc. The increase in skin temperature can improve vasodilation and microcirculation as well as exert antibacterial effects.^{187–189} Kang et al. developed a graphene-based thermal patch with targeted thermotherapy. The thermotherapy was performed via a graphene-based heater, which can rapidly reach 40°C in 20 s and maintains 32°C for accelerated healing in rat wound models via vasodilation-enhanced oxygen/nutrient delivery (Figure 6A).¹⁵

Chronic wounds are often accompanied by bacterial infections. Photothermal therapy can use light and photothermal conversion agents to convert light energy into heat energy, improving microcirculation, regulating the microenvironment, and killing bacteria. Common photothermal agents include noble metal (e.g., gold NPs) and metal oxide, MXene, reduced GO (rGO), etc.¹⁹³ Ma et al. developed an NIR-responsive smart bandage with rGO/polyacrylamide hydrogel serving as a photothermal agent (Figure 6B). The gel showed good photostability (Figure 6C). The smart bandage can rapidly heat to ~50°C under NIR irradiation to kill *S. aureus* and achieved 89.2% wound closure with complete re-epithelialization on the seventh day in infected mouse models.⁹⁰ Photothermal therapy can also be combined with CDT or ES to achieve synergistic and enhanced treatment.^{28,194}

There are still some challenges. First, the precision and uniformity of temperature control remain suboptimal; systems that lack real-time closed-loop feedback may result in thermal damage.^{15,28} Second, the penetration depth is limited, resulting in poor efficacy for deep or thick-walled wounds.^{28,90,193} Third, the synergistic mechanisms between thermotherapy and other treatment modalities have not yet been fully elucidated, hindering the optimization of combination therapies.

Ultrasound treatment

Ultrasound therapy, a kind of physical energy therapy, exhibits advantages of non-invasiveness, strong penetrability, and spatiotemporal controllability, providing a critical tool for promoting wound healing.¹⁹⁵ The reverse piezoelectric effect of lead zirconate titanate (PZT)-4 is often used to generate ultrasonic waves under voltage excitation. Lyu et al. developed a flexible ultrasonic patch integrating discretely arranged PZT-4 piezoelectric ceramic units on a flexible substrate with an “island-bridge” metal electrode structure, featuring excellent skin conformability and mechanical stability (Figure 6D). A thin



(legend on next page)

hydrogel layer acts as both encapsulation and acoustic coupling medium, ensuring efficient ultrasound penetration. The flexible design enables adjustable sound field focusing (Figure 6E). The low-intensity pulsed ultrasound can activate Rac1 in both dermal and epidermal layers, promoting the reversal of fibroblast senescence, accelerating angiogenesis, and re-epithelialization. *In vivo* chronic diabetic wound models have demonstrated that daily 20-min ultrasound treatment shortens wound healing time.¹⁶⁶ In addition, the positive piezoelectric effect of PZT-4 can also be utilized. Chen et al. designed an ultrasonic-electric-coupled patch (UEP) integrating PZT-4 piezoelectric arrays, energy exchange modules, and electric stimulation coils (Figure 6F). The patches utilize PZT-4's reverse piezoelectric effect to generate focused ultrasound for deep tissue stimulation, while excess ultrasonic energy is converted to pulsed ES via the positive piezoelectric effect.¹⁹⁰

Ultrasound therapy offers non-invasiveness, deep tissue penetration, and spatiotemporal controllability for wound healing, yet several challenges remain. First, traditional rigid transducers lack flexibility, limiting conformal contact with irregular wounds and stable acoustic coupling. Second, the biological mechanisms and parameter-dependent effects of ultrasound on different cell types remain poorly understood, and optimal parameters for various wounds have not been established.^{166,190} Third, the integration of ultrasound with other modalities in closed-loop systems, as well as their synergistic mechanisms, remains underexplored.¹⁹⁰

Other treatment

Biofilm can offer advantages for pathogen survival, raising the effective minimum inhibitory concentration of several antibiotics against pathogens and making it an important factor contributing to microbial persistence in wounds. A potential solution is the use of antimicrobial molecules. Bozyel et al. developed an electrochemical bandage enabled by programmable multimodal wearable potentiostats to achieve dual-action therapy via sequential generation of HOCl and H₂O₂. The bandage generates HOCl for potent bactericidal activity, eliminating biofilms of MRSA, *P. aeruginosa*, and *Candida auris*, and H₂O₂ to promote healing. The treatment program reduces pathogen loads *in vitro* and in MRSA-infected murine wounds.¹⁹⁶ UV radiation can also be used due to its inherent antimicrobial property of damaging DNA. Ullah et al. integrated UVC LEDs into a smart bandage, leveraging UV radiation to eradicate Gram-negative bacteria within 6 h, reducing biofilm formation to <30% surface coverage (vs. 60% for controls).¹⁹⁷

Among the issues hampering chronic wound healing, suboptimal oxygenation of is one of the most critical. Ochoa et al. designed paper-based wound dressings integrating O₂ sensing and delivery. The dressings generate O₂ via manganese dioxide-catalyzed decomposition of H₂O₂, increasing wound bed oxygen levels by 13% in 1 h and covering a 10 mm radius area. The platform doubles, or triples, wound oxygen to clinically relevant levels *in vivo*. However, it was observed that oxygenated wounds healed slightly slower than controls, presumably due

Figure 6. Thermal and ultrasound therapy

- (A) Temperature profiles of graphene-based heaters at an input voltage of 5 V. Reproduced from Kang et al.¹⁵ under a CC BY-NC 4.0 license.
- (B) IR camera images of rGO/PAM gel containing 0, 0.1, 0.2, and 0.4 mg mL⁻¹ rGO W/NIR.
- (C) Photothermal stability of rGO/PAM gel (0.2 mg mL⁻¹ rGO) with three heating/cooling cycles in W/NIR. Reproduced, with permission, from Ma et al.⁹⁰ Copyright 2025, American Chemical Society.
- (D) The flexible ultrasonic patch can completely conform to the skin (bending radius <5 mm).
- (E) Radiation sound field in the treatment area when the ultrasonic patch is under different bending conditions. Sound waves are distributed vertically along the direction of the array when the bending degree is 0°. Sound waves are focused on the center of the bending circle when the bending degree is 15°. Reproduced, with permission, from Lyu et al.¹⁶⁶ Copyright 2021, Wiley.
- (F) Schematic of the structural layout of UEP. Reproduced, with permission, from Chen et al.¹⁹⁰ Copyright 2025, Wiley.
- (G) Schematic depicting iSAFE as a multi-functional wound electronic capable of closed-loop sensing and treatment for infected chronic wounds. A highly integrated configuration endows iSAFE with superior breathability, waterproofness, tissue adhesiveness, anti-bacterial activity, and on-demand therapeutic ability. Reproduced from Huang et al.¹⁴¹ under a CC BY-NC-ND 4.0 license.
- (H) Left: Schematic of the working mechanism of B-TENG. Right: Output voltage of B-TENG with different borophene concentrations. Inset: the output comparison between an optimal (20 wt %) and pristine device (0 wt %). Reproduced, with permission, from Chen et al.¹⁹¹ Copyright 2022, Wiley.
- (I) TGC dressing generates an exogenous electric field (EF) that facilitates wound healing by guiding and expediting the migration of epithelial cells. Tc and Th represent the temperature of the cold and hot side, respectively. φ denotes the electric potential.
- (J) Left, The initial potential difference between the wound bed and the edge. Right: The potential difference tested with TGC dressing. Reproduced, with permission, from Xin et al.¹⁹² Copyright 2025, Springer Nature.
- (K) Magnified image of WPED showing the water-powered battery.
- (L) Exploded-view schematic of WPED showing the main components of the system. Reproduced from Kaveti et al.¹⁶⁵ under a CC BY-NC 4.0 license.
- (M) Schematic of the energy storage mechanism of the ACZB.
- (N) Ten cycles of air self-charging for 2 h/galvanostatic discharging at 50 μ A cm⁻². Reproduced, with permission, from Zhou et al.¹⁸⁶ Copyright 2024, American Chemical Society.
- (O and P) Wound classification (O) and healing time (P) accuracy based on different machine learning models and features used. Features in (O) progress from H₂O₂ only to combinations with pH, diabetes, dialysis, PAD, and age. Features in (P) progress from NO only to combination with pH, wound area, PAD, age, and dialysis. Error bars represent the SD of the mean from three machine learning (ML) results. SVC, support vector classification; Rbf, radial basis; KNN, k-nearest neighbor algorithm.
- (Q) Shapley additive explanations (SHAP) decision plot explaining how the ML model determines wound classification (left) and healing time (right) using combinations of biochemical and patient features. Color scale, low (blue) to high (magenta). For wound classification (left), the outputs 1, 2, 3, and 4 represent Wagner grade 1, Wagner grade 2, Wagner grade 3, and clinical-etiology-anatomy-pathophysiology (CEAP) C6, respectively. For wound healing time prediction (right), 1, 2, 3, and 4 represent less than 1 month, from 2 to 5 months, from 6 to 9 months, and over 10 months, respectively. From Wang et al.²⁶ Reprinted with permission from AAAS.

to H_2O_2 leakage, highlighting the need to optimize pumping accuracy and sealing design to mitigate unintended tissue interaction.^{105,198}

Apart from photothermal therapy, light can also be combined with photosensitizers for photodynamic therapy. Xue et al. developed an optoelectronic patch integrating photodynamic and light therapy. Paired with PVA/hyaluronic acid microneedles loaded with photosensitizer, the patch features a circuit with a violet LED for photosensitizer activation and four red LEDs to reduce inflammation and promote angiogenesis and collagen deposition.¹⁶⁷

INTEGRATED CLOSED-LOOP SYSTEMS

An integrated closed-loop system for wound management represents the current highest-level autonomous paradigm of intelligent wound care, which operates as a complete and self-sustaining cycle: real-time sensing → (multi-parameter) data integration and transmission → AI analysis → draw conclusions and/or make decisions → therapeutic intervention → dynamic feedback regulation. It tightly integrates power management, information acquisition, data transmission and analysis, controllable treatment systems, and remote control, which can respond to dynamic changes, optimize care protocols, and enhance treatment precision and timeliness. Rather than a simple assembly of separate modules, a genuine closed-loop system emphasizes synergistic interaction among all components to achieve autonomous, adaptive, and personalized therapy.

Sensing, data integration, and transmission

Sensing and data integration constitute the input foundation of the entire closed-loop system, providing real-time and comprehensive information for subsequent analysis and decision-making.

Compared with single-parameter sensing, multi-parameter sensing and combination can help to enable more comprehensive monitoring and earlier and more specific identification of wound states, improving the accuracy of diagnosis.¹⁹⁹ For example, by simultaneously measuring O_2 , UA, lactate, pH, and temperature, it was observed that there is a strong negative correlation between O_2 and other biomarkers, and there were statistically significant differences in these parameters between infected and uninfected wounds.¹⁴ In addition, as inductance-capacitance resonators produce only a single and weak spectral resonance, these systems unavoidably suffer from limited sensitivity and sensing capacity, impeding the simultaneous detection of multiple minute physiological variations. To address this, Ye et al. developed a multiplexed sensing system based on a third-order exceptional point in non-Hermitian physics, aiming to enhance sensitivity and enable multi-parameter detection. The system features three distinct resonance frequencies, allowing a single resonator to simultaneously detect resistive and capacitive biomarkers, and achieving over 10-fold sensitivity improvement compared to conventional resonator-based systems.²⁰⁰

In wearable wound management devices, communication modules are essential for transmitting data and treatment control signals in real time between the device and external platforms,

enabling remote monitoring and timely intervention. Bluetooth,^{14,15,20} near-field communication (NFC),^{16,21,26,29} and WiFi⁶³ are often used for signal transmission. Reliable wireless communication ensures rapid and stable information flow within the closed loop, supporting timely responses to dynamic wound alterations.

AI assistance

Traditional wound assessment methods, such as intermittent visual inspection and time-consuming culture techniques, mainly depend on subjective visual inspection and time-consuming off-site testing, which fail to accurately evaluate the conditions of wounds and perform intervention in time. Advanced data analysis and interpretation tools are required to process the large amount of information harvested from bioelectronics. AI, acting as the core decision-making unit in the feedback loop, can assist in real-time analysis and calibration of sensing data to realize wound classification, evaluation, and prediction and translate raw sensing data into actionable commands, such as issuing infection alerts or initiating therapeutic interventions.

During the sensing process, some parameters may be affected by interference from other parameters. Therefore, calibration in real time is necessary to enhance the data accuracy for wound status evaluation. Kaewpradub et al. employed partial least squares (PLS) regression to address the inaccuracy of traditional linear calibration in PYO detection caused by dynamic pH fluctuations in wound exudate. As PYO and its electrochemical response are pH dependent, the lack of pH correction may lead to significant measurement deviations. The PLS model integrates 41 SWV current data points from the PYO sensor and 3 potentiometric data points from the PANI/CNT pH sensor, establishing a predictive relationship between multivariate signals and actual PYO concentrations, achieving high accuracy and recovery rates, effectively mitigating the inaccuracy problem of traditional methods, and enhancing the reliability of sensors for real-time monitoring.²⁰¹

To eliminate reliance on subjective evaluations or historical clinical data, algorithms can be trained to enable objective wound management by integrating multimodal data. AI algorithm models are commonly used for wound classification, evaluation, and prediction. Through Shapley additive explanation analysis of the feature relevance, the authors determined that diabetes, O_2 , and H_2O_2 played the most important roles in classifying wounds, whereas peripheral artery disease, wound area, pH, and NO were important for predicting healing time, achieving classification accuracies of 88.75% and 94.0% respectively, using a support vector classification linear model (Figure 6O–6Q). Features such as NO, H_2O_2 , diabetes, and age were also used to estimate wound depth, with an accuracy of 81.4%.²⁰ Similarly, Wang et al. employed a deep neural network (DNN) for wound state classification, integrating multimodal sensor data such as temperature, pH, UA, and glucose, which can achieve 91% accuracy in distinguishing inflammatory from normal healing states. Moreover, this AI-driven framework can directly trigger electrolysis-driven microfluidic drug release in closed-loop feedback, enabling autonomous and timely therapeutic responses without manual intervention.¹⁴² Additionally, the convolutional neural network (CNN) model can be used to analyze RGB signals

converted from colorimetric pH-sensing hydrogel to automatically evaluate wound states, achieving an accuracy of 94.47%.²⁰² A support vector machine model trained on these frequency-domain features of lactate concentration time-series data in the first 3 days post injury could predict wound closure rate with ~76% accuracy, which improved to ~83% when integrating pH.²¹ Wound resistance dynamically correlates with the healing stage. Chen et al. employed the random forest machine learning algorithm to analyze wound resistance data in order to classify healing stages and predict healing time, achieving a high test accuracy of 96.66%, enabling noninvasive and real-time monitoring of wound status and precise forecasting of wound recovery.^{190,203}

AI can also integrate heterogeneous data to overcome the limitations of a single technology. Sun et al. utilized machine learning-based information fusion to integrate odor data from an electronic nose and FAIMS technology using a three-module framework of feature extraction, feature selection, and model fusion using different algorithms, achieving an accuracy of 94.63% and 88.05% for 24 and 48 h infection in distinguishing three bacterial infections from non-infected wounds, outperforming single-technology models with enhanced sensitivity and specificity.¹⁴⁶

Power supply

A stable and sustainable power supply is the essential prerequisite for the operation of bioelectronic systems, ensuring that sensing, analysis, communication, and treatment modules work synchronously over long periods. Batteries are a mature and reliable solution, that has the advantages of stable and reliable output and easy integration and is not limited by external conditions. A disadvantage is the contradiction between battery capacity and duration as well as the comfort issues brought by rigid batteries. A battery-free design eliminates the need for battery replacement and charging, which is expected to achieve a long-term power supply, improving patient comfort. However, disadvantages may be unstable energy output, low power density, and high dependence on the external environment.

A battery-free power supply aims to collect energy from the surrounding environment. Power can be collected from an inductive coil (including NFC),^{16,29,197} the triboelectric effect,¹⁹¹ temperature gradients,¹⁹² redox reactions of electrode materials,^{165,186} biofuel cells,¹⁸⁵ etc. A battery-free power supply based on an inductive coil has emerged as a common contactless power supply strategy. The system typically comprises a primary coil integrated in an external power transmitter and a secondary coil embedded in the flexible wearable device, leveraging resonant magnetic induction to transfer energy. When an alternating current passes through the primary coil, it generates an oscillating magnetic field to induce an electromotive force in the secondary coil, enabling energy conversion. While these methods require that patients are in the vicinity of a transmitting antenna to wirelessly power the device, this restricts patient mobility. In addition, these systems also require meticulously designed transmission and receiving antennas and rely on expensive and bulky equipment.

Triboelectric nanogenerators (TEGs) can harvest mechanical energy to generate power. For instance, Chen et al. developed a self-powered fabric-based TENG (B-TENG) based on a boro-

phene/ecoflex nanocomposite and functional fabric for continuous wound monitoring and ES therapy. The TENG harvests ambient mechanical energy via the triboelectric effect, generating a maximum output of 120 V and 20 μ A by contact-separation between the borophene/ecoflex negative layer and polyester positive layer (Figure 6H).¹⁹¹ However, due to the fact that the triboelectric effect requires body movement or external mechanical stimulation to generate electrical energy, there may be issues with unstable output.

To address the reliance on external power sources, Xin et al. developed a self-powered thermogalvanic cell (TGC) dressing that harvests energy from the temperature gradient between the wound and the ambient environment. Composed of $\text{Fe}^{2+}/\text{Fe}^{3+}$ cross-linked alginate hydrogel reinforced with nanofibers (Figure 6I), the TGC dressing converts thermal energy into electrical energy via the thermogalvanic effect, generating a thermopower of 1.7–3.6 mV K^{-1} and an ionic conductivity of 1.84 S m^{-1} at room temperature. When applied to wounds, the temperature difference induces an exogenous electric field up to ~53 mV enhancement of wound potential difference (Figure 6J). This self-sustaining power supply not only drives ES for wound healing but also supports a wireless monitoring system to track wound temperature and exudate-related signals.¹⁹²

The essence of a chemical battery lies in the redox reactions of the electrode materials. Through the directional transfer of electrons, chemical energy is converted into electrical energy. To address the limitations of bulky external power sources in wound electrotherapy, Kaveti et al. developed a water-powered magnesium-silver/silver chloride (Mg-Ag/AgCl) battery, consisting of a Mg anode, Ag/AgCl cathode, and NaCl-loaded dry cellulose separator (Figures 6K and 6L). Upon activation with a small volume of water or wound exudate, the battery generates an initial output voltage of 1.5 V, which gradually decreases until reaching 0 V. This water-triggered power supply eliminates the need for pre-charging or external electronics, offering on-demand energy for wound electrotherapy.¹⁶⁵ Similarly, based on the redox reactions, Zhou et al. developed an air self-charging Zn battery (ACZB) for infected wound care, integrating a zinc anode, viologen-based dual-network hydrogel cathode, and gelatin/ ZnSO_4 gel electrolyte. The battery generates electricity via spontaneous redox reactions of Zn oxidation and O_2 reduction, with discharged viologen (V^+) spontaneously reoxidized to V^{2+} by ambient oxygen for rapid self-charging (Figure 6M). Operating at open-circuit voltages in the range of 0.95–1.10 V, this battery can maintain 51.8% of its initial capacity after 10 charge-discharge cycles (Figure 6N) and supply power for therapeutic ES while releasing bioactive Zn^{2+} without the need for external power sources.¹⁸⁶

Rezaie et al. developed a wearable microbial fuel cell as a self-powered source for wound care, leveraging *Bacillus subtilis* endospores decorated with biosynthesized SnO_2/CuO NPs as biocatalysts. Fabricated on a flexible paper substrate integrated with conductive graphene-alginate-gelatin hydrogel, the microbial fuel cell is activated by nutrient-rich wound exudate, which triggers endospore germination to generate electricity and antibacterial agents. Its horizontal configuration with the anode on the wound and the cathode on healthy skin uses skin as an external circuit, enabling stable power output while delivering

ES.¹⁸⁵ An enzyme-powered biofuel cell, such as a lactate oxidase/Ag₂O system, can also be used to power sensors and electronic components.²¹

System-level integration

System-level integration is the essence of closed-loop systems. The defining feature of an integrated closed-loop system is a synergistic integration of all modules rather than independent operation. In a fully closed-loop workflow, (1) multimodal sensing provides real-time data input; (2) AI processes and analyze data, then draws conclusions and/or makes decisions, such as classification, evaluation, prediction, warning and/or intervention; (3) treatment is started or stopped; and (4) power supply and wireless transmission technologies guarantee long-term, untethered operation. For example, Huang et al. developed an NFC-powered, fully permeable, skin-adhesive, stretchable electronic wound bandage (iSAFE) integrating *in situ* sensing, wireless communication, and adaptive drug delivery. Equipped with stretchable sensors for real-time monitoring of glucose, pH, and temperature, the system transmits data to smartphones via NFC for analysis. Triggered by abnormal biomarker signals indicating infection or delayed healing, voltage-modulated electrodes can enable localized drug release via iontophoresis, with the release amount precisely controlled by voltage and duration (Figure 6G). In *E. coli*-infected diabetic rat models, the closed-loop system accelerated wound closure, reduced inflammatory factors, and enhanced angiogenesis and collagen deposition. Clinical trials on 10 patients further validated its ability to correlate sensor data with wound bed scores, guiding personalized treatment. With superior biocompatibility, stretchability, permeability, and waterproofness, this integrated platform exemplifies a practical closed-loop paradigm for intelligent wound care.¹⁴¹

The iSAFE platform and other representative demonstrations confirm that such an integrated design enables autonomous adaptation to dynamic wound microenvironments, automatically optimizes therapeutic schemes, and significantly improves the precision and efficiency of wound care. Future closed-loop systems will continue to evolve toward interpretable AI, multimodal synergistic therapy, biodegradable materials, and large-scale clinical validation, ultimately realizing next-generation intelligent, home use, and clinically translatable wound management platforms.

Integrated closed-loop systems represent the ultimate paradigm for intelligent wound management, but their development remains at an early stage, facing multiple integration challenges. First, compatibility and crosstalk between multimodal sensing and therapeutic modules pose significant hurdles. Integrating temperature, pH, multiple biochemical sensors, and ES or drug release units within a single platform imposes stringent requirements on materials selection, circuit design, and fabrication processes and may lead to signal crosstalk, electrochemical interference, and mechanical coupling between sensors and actuators.^{141,177} Second, the long-term stability and biocompatibility of power supply systems require improvement. Although wireless power transfer (e.g., NFC and radiofrequency) eliminates battery replacement, its limited operating distance restricts patient mobility. Self-powered systems (e.g., triboelectric, thermoelectric, and biofuel cells) are conceptually appealing, but their low power density and susceptibility to environmental conditions make it

difficult to meet the stable power demands of complex systems.^{16,29,165,185,186} Third, the clinical interpretability and generalizability of AI-assisted decision-making remain to be validated. Most existing studies employ machine learning models trained on small datasets, and their generalizability across different centers and populations has not been adequately demonstrated. Moreover, AI model outputs often lack biological interpretability, limiting clinical acceptance and trust.^{20,21} Future efforts should focus on interdisciplinary collaboration to develop highly integrated, flexible, hybrid electronic systems, optimizing electromagnetic compatibility and mechanical matching among multiple modules. Concurrently, multi-center, prospective clinical cohort studies should be promoted to construct high-quality, standardized multimodal wound databases, providing a solid foundation for training and validating AI models. At the algorithmic level, explainable artificial intelligence (XAI) methods should be explored to enhance the transparency of model decision-making processes, improving clinical acceptance.

SUMMARY AND PROSPECTS

In summary, we reviewed advancements in flexible bioelectronics and smart wearable devices in the field of wound management, covering topics ranging from exudate management and multimodal sensing to advanced therapeutic interventions and fully integrated closed-loop systems. The field is rapidly evolving from passive monitoring tools to autonomous platforms capable of real-time diagnosis and on-demand treatment. Although significant progress has been made, substantial efforts are still required to overcome technical and ethical challenges before successful translation into clinical practice can be achieved.

Relatively mature directions

Microfluidic technologies for collecting wound exudate have made significant strides, and the application of biomimetic design in optimizing fluid management is becoming increasingly common. Electrochemical and optical sensing strategies are well established for monitoring physical and small-molecule biochemical markers. In terms of treatment, drug-free modalities are increasingly integrated into wearable platforms. Wireless power supply and flexible structural design not only enhance the portability of the device but also improve its wearability and mechanical stability.

Challenges need to be addressed

Although significant progress has been made in this field, several technical challenges remain. First, biosensors require a sufficient amount of wound exudate to function properly; however, chronic wounds produce very little exudate, and the rate and volume of exudate production cannot be controlled manually. These factors can lead to variations in the concentration of collected wound exudate, affecting accuracy. Additionally, because wound exudate contains a variety of complex substances, these substances may interfere with the accurate detection of certain target analytes. Second, most existing devices are capable of monitoring physical parameters as well as a few conventional analytes continuously. However, wound exudate also contains

a large number of substances—including proteins, enzymes, and lipids—that hold value for early diagnosis, but, long-term, continuous detection of these biomarkers is still a challenge.

In addition to technical challenges, there are also ethical concerns that need to be solved. First, issues related to data privacy and cybersecurity in smart wound dressings cannot be overlooked. These devices continuously collect various physiological data, biochemical indicators, and even image data from wounds, all of which constitute highly sensitive personal health information. There is a risk that these data could be stolen or leaked during wireless transmission or upload to cloud storage. Second, it is difficult for clinicians to trace the decision-making process of proprietary AI algorithms. Consequently, when an AI system makes an erroneous decision that results in patient harm, determining liability becomes a complex ethical issue. Furthermore, the accuracy of AI models is highly dependent on the quality and representativeness of training data. If training data fail to adequately cover wound characteristics across patients of different races, skin tones, ages, or with comorbidities, the AI system may exhibit algorithmic bias, leading to inaccurate diagnoses or treatments for specific patient groups. Most importantly, AI may autonomously make treatment decisions without human intervention in closed-loop systems, which complicates the informed consent process. The boundary between the degree of autonomy of AI and human intervention needs to be clear.

Realistic translational opportunities in the next 3–5 years

From a technical perspective, a more feasible near-term goal is not to pursue a fully autonomous closed-loop control system but rather to focus on integrating thoroughly validated physical and biochemical sensors into flexible patches or dressings to enable long-term, quantitative monitoring of key indicators such as pH, temperature, impedance, and glucose levels. Such platforms synchronize data to clinical terminals via encrypted wireless transmission, providing clinicians with objective, real-time insights into wound status, while treatment decisions remain under the clinician's control. This approach not only significantly lowers regulatory barriers and algorithmic risks but also facilitates the collection of high-quality, multicenter data in real-world clinical settings, laying the groundwork for the subsequent training and validation of closed-loop algorithms.

In terms of regulatory and ethics coordination, researchers and industry stakeholders should proactively engage with regulatory authorities early in the product development process to clarify the classification of smart wound management devices, the design of clinical endpoints, and post-market regulatory requirements. Concurrently, medical ethicists, legal experts, and engineering teams should collaborate to develop ethical guidelines for smart wound dressings, clarifying the specific requirements for informed consent, the oversight hierarchy for human-machine collaboration, and mechanisms for detecting and correcting algorithmic bias. Furthermore, conducting rigorous health economic evaluations to demonstrate the overall cost-effectiveness—achieved through reduced infection-related hospital stays, lower amputation rates, and shorter care cycles—will serve as a key lever for driving health insurance coverage and market acceptance. Driven by the synergy of technology, regu-

lation, ethics and economics, the next generation of smart wound dressings is poised to make a genuine leap from laboratory innovation to accessible clinical solutions in the future.

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AUTHOR CONTRIBUTIONS

Writing – original draft, H.F. and F.Q.; writing – review and editing, Z.C., J.J., and B.L.; conceptualization and funding acquisition, Yanhua Wang, E.H., Yilin Wang, and D.Z.; visualization, C.X., S.G., and Z.Y. All authors have read and agreed to the published version of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests

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